

# Chromatin accessibility variation provides insights into missing regulation underlying immune-mediated diseases

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## eLife Assessment

This paper addresses a significant question regarding the low overlap between genetic discoveries for human complex diseases and those for gene expression by emphasizing the contribution of cell-type-specific chromatin accessibility QTLs. The analyses supporting the main claims are **convincing**, and the key conclusions are **valuable** and of interest to readers in the fields of human genetics and functional genomics.

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## Abstract

Most genetic loci associated with complex traits and diseases through genome-wide association studies (GWAS) are noncoding, suggesting that the causal variants likely have gene regulatory effects. However, only a small number of loci have been linked to expression quantitative trait loci (eQTLs) detected currently. To better understand the potential reasons for many trait-associated loci lacking eQTL colocalization, we investigated whether chromatin accessibility QTLs (caQTLs) in lymphoblastoid cell lines (LCLs) explain immune-mediated disease associations that eQTLs in LCLs did not. The power to detect caQTLs was greater than that of eQTLs and was less affected by the distance from the transcription start site of the associated gene. Meta-analyzing LCL eQTL data to increase the sample size to over a thousand led to additional loci with eQTL colocalization, demonstrating that insufficient statistical power is still likely to be a factor. Moreover, further eQTL colocalization loci were uncovered by surveying eQTLs of other immune cell types. Altogether, insufficient power and context-specificity of eQTLs both contribute to the 'missing regulation.'

## Introduction

More than a decade of genome-wide association studies (GWAS) has revealed several properties of the genetic architecture of complex traits and diseases<sup>1,2</sup>. Most (~93%) of the genetic associations are detected in the noncoding portion of the genome<sup>3</sup>, and disease heritability is concentrated in putative regulatory regions<sup>4</sup>. Expression quantitative trait loci (eQTLs), which are loci associated with gene expression levels, are enriched for trait associations<sup>5,6</sup>. Complex traits are also characterized by their extreme polygenicity, where individual genetic association has only a small effect on the trait<sup>7</sup>. Altogether, these observations have led to a prevalent theory that causal genetic variants affect regulation of key genes across the genome, where each gene explains a modest proportion of trait variation<sup>8</sup>. There are experimental strategies aimed at nominating putative causal genes at noncoding GWAS loci<sup>9–11</sup>. As an alternate approach, an eQTL signal colocalizing with the GWAS signal illustrates the effect of the causal variant on gene expression and suggests that the affected gene contributes to the trait. Detection of disease-associated eQTLs thus can identify putative disease genes, helping to elucidate disease mechanisms and develop therapeutics targeting them<sup>12</sup>.

Although it has become expected that eQTLs will be discovered in most noncoding GWAS loci, only a minority of trait-associated loci have been explained by eQTLs<sup>13–16</sup>. The Genotype-Tissue Expression (GTEx) eQTL study across 49 human tissues recognized that, for a typical complex trait, about 20% of GWAS loci contained a colocalized eQTL in the *cis* region (*i.e.*, 1 Mb) around the gene (*i.e.*, *cis*-eQTL)<sup>14</sup>. Even when focusing just on putatively causal genes, the rate of colocalization was very low (8%)<sup>15</sup>. Furthermore, the proportion of trait heritability mediated by *cis*-eQTLs ( $h^2_{\text{med}}/h^2_{\text{SNP}}$ ) of assayed gene expression was estimated to be only about 11% on average<sup>16</sup>. We will call the missing link between genetic association to traits and regulatory function of the associated noncoding variants as ‘missing regulation’, as Connally and colleagues introduced<sup>15</sup>. To be able to detect eQTLs in the unexplained disease-associated loci, a better understanding of the possible reasons for why they have been missing is essential.

There are many possible explanations for why disease-associated loci are missing colocalized eQTLs<sup>15,17–19</sup> and for why  $h^2_{\text{med}}/h^2_{\text{SNP}}$  estimates for eQTLs are relatively low<sup>16</sup>. First, statistical power to detect disease-associated eQTLs may be insufficient<sup>19</sup>. For example, negative selection against gene expression variation may lead to challenges in detecting eQTLs for trait-relevant genes<sup>18,20</sup>. Since disease genes are likely to be dosage sensitive, there would be selection against their having large eQTL effects<sup>20</sup>. Consequently, the negative selection induces a “flattening” effect, in which weak eQTL variants, often in regions distal to the gene’s promoter<sup>21</sup> may reach high enough frequency, whereas strong eQTL variants remain at low frequency<sup>7</sup>. In fact, eQTL-mediated heritability was enriched in genes showing mutational constraint and those with lower *cis*-heritability<sup>16</sup>. These weaker or low-allele-frequency eQTL effects would require larger sample sizes to be detected with statistical significance and to show colocalization<sup>19</sup>. Second, causal eQTL effects may be specific to cell types that have not been assayed<sup>17</sup>. For example, immune cell eQTLs are highly cell-type specific, and eQTL effects specific to some immune cell types may mediate immune disease risk<sup>22</sup>. Specificity of eQTL effects can also be limited to specific cell states<sup>23,24</sup>. Detecting cell-type or cell-state specific eQTL effects requires the necessary gene expression data sets from the relevant cell types and states<sup>17</sup>, which has been a limiting resource for such analyses.

To investigate why disease-associated eQTL signals have been missing, we focused on immune-mediate diseases (IMDs) as a model set of complex traits. We aimed to collect IMD-associated loci that are expected to show eQTL signals in some cell type. Since active regulatory elements coordinate target gene expression<sup>25</sup>, we reasoned that variants that affect chromatin phenotypes at regulatory elements, such as transcription factor (TF) binding<sup>26–28</sup> and

chromatin accessibility<sup>29,30</sup>, have the potential to impact gene expression<sup>31</sup>. These chromatin phenotypes may show detectable genetic effects even when an eQTL effect in the same cell type was not identified in the locus<sup>32</sup>. For example, only about 20% of lymphoblastoid cell lines' (LCLs') PU.1 binding QTLs (bQTLs) that colocalized with blood cell traits' association showed an eQTL effect for a nearby gene in LCLs<sup>33</sup>.

Here, we analyzed genetic and functional genomic (*i.e.*, ATAC-seq and RNA-seq) data in LCLs. LCLs are derived from B lymphocytes, and their *cis*-regulatory elements were enriched for variants associated with some IMDs<sup>34,35</sup>. We evaluated whether chromatin accessibility QTLs (caQTLs) in LCLs potentially explain IMD associations using mediated heritability analysis<sup>16</sup> and colocalization<sup>36,37</sup>. Then, we searched for disease-associated loci that were significant caQTLs, but not eQTLs.

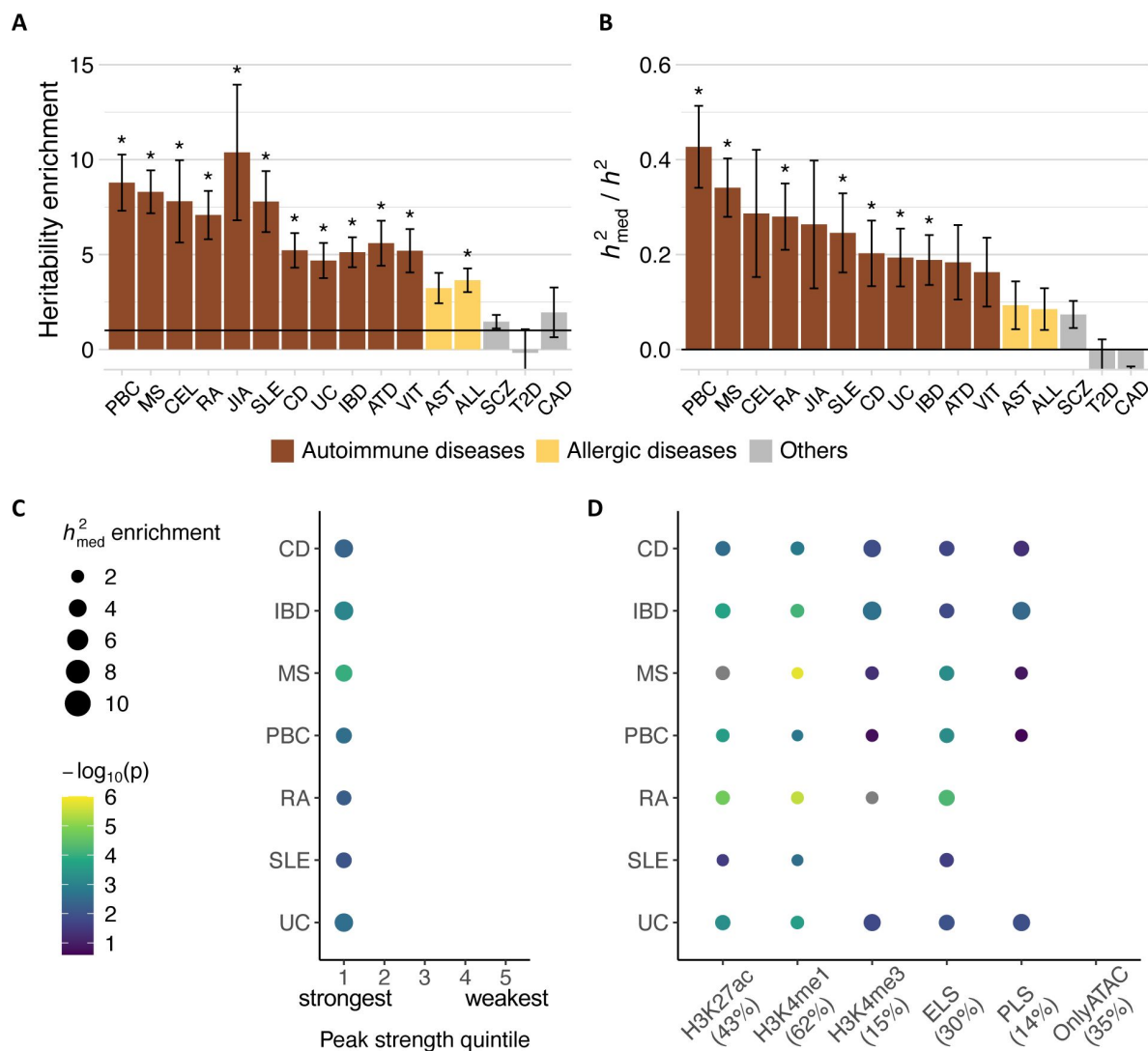
We examined whether the various potential reasons for missing eQTLs can account for IMD-associated loci that are explained by caQTLs but not eQTLs. First, we explored the extent to which eQTLs may have been missed because of limited statistical power. We compared *cis*-heritability of colocalized caQTLs and eQTLs stratified by distance between the accessible region and the transcription start site of the associated gene. We also investigated whether meta-analysis of published LCL eQTL summary statistics, in order to effectively increase the sample size, can uncover previously missed eQTLs. Second, we surveyed whether cell-type specificity of regulatory variant effect may account for the missing regulation. We surveyed various immune cell eQTL data to identify loci with which they colocalize even if LCL eQTLs did not colocalize with those loci.

Through this study, we present how regulatory QTLs beyond eQTLs, such as caQTLs, can be effective in detecting the potential molecular consequences of disease-associated variants. Moreover, results from inspecting disease-associated loci where genetic effects are detected on chromatin accessibility but not on expression suggest reasons why the effects on gene expression may have been missed. These results provide insights on which strategies may be effective in uncovering more genes that underlie diseases.

## Results

### Accessible chromatin in LCLs explains a significant proportion of immune-mediated disease heritability

We aimed to evaluate whether variants that alter chromatin accessibility in LCLs may explain genetic associations to IMD. First, we verified whether accessible regions in LCLs are enriched for IMD heritability. We reanalyzed 100 LCL ATAC-seq samples<sup>30</sup> to define accessible regions in this cell type. With stratified LD score regression (S-LDSC)<sup>38</sup>, we estimated their heritability enrichment across 13 IMDs, including 11 autoimmune diseases — autoimmune thyroid disease (ATD)<sup>39</sup>, celiac disease (CEL)<sup>40</sup>, Crohn's disease (CD)<sup>41</sup>, inflammatory bowel disease (IBD)<sup>41</sup>, juvenile idiopathic arthritis (JIA)<sup>42</sup>, multiple sclerosis (MS)<sup>43</sup>, primary biliary cholangitis (PBC)<sup>39</sup>, rheumatoid arthritis (RA)<sup>44</sup>, systemic lupus erythematosus (SLE)<sup>45</sup>, ulcerative colitis (UC)<sup>41</sup>, and vitiligo (VIT)<sup>46</sup> — and 2 allergic diseases — allergy (ALL)<sup>47</sup> and asthma (AST)<sup>47</sup>. We also analyzed GWAS data for 3 non-immune diseases — type 2 diabetes (T2D)<sup>48</sup>, coronary artery disease (CAD)<sup>49</sup> and schizophrenia (SCZ)<sup>50</sup> — for comparison. Single nucleotide polymorphisms (SNPs) in accessible regions in LCLs were significantly enriched for IMD heritability ( $p < 0.003125$  (Bonferroni-corrected), S-LDSC; **Figure 1A**) and there was no significant enrichment for non-immune diseases ( $p > 0.05$ , S-LDSC). These results indicate that accessible regions in LCLs harbor many variants specifically associated with IMDs, and therefore that LCLs share IMD-associated accessible regions with those of the causal cell type(s).



**Figure 1.**

### Immune disease heritability mediated by chromatin accessibility in LCLs.

(A-B) Heritability enrichment in accessible regions in LCLs, based on (A) stratified LD score regression (S-LDSC) and (B) the proportion of heritability mediated by caQTLs in LCLs based on mediated expression score regression (MESR). For both, error bars represent jackknife standard errors of the mean. The color of the bars indicates disease type. \*:  $p < 0.003125$  (Bonferroni-corrected).

(C) Mediated heritability enrichment of accessible regions by peak strength quintile. The strongest peaks are in the 1st quintile, and the weakest peaks are in the 5th quintile. Only enrichment values with FDR < 5% (based on q-value) are shown. Stronger and more significant enrichment indicate that mediated heritability is concentrated in that subset. (D) Mediated heritability enrichment of accessible regions by histone mark annotation. The percentages in parentheses represent the proportion of accessible regions with the indicated histone mark. Only enrichment values with FDR < 5% (based on q-value) are shown. Color and size of the points are on the same scale as in (C). PBC, primary biliary cholangitis; MS, multiple sclerosis; CEL, celiac disease; RA, rheumatoid arthritis; JIA, juvenile idiopathic arthritis; SLE, systemic lupus erythematosus; CD, Crohn's disease; UC, ulcerative colitis; IBD, inflammatory bowel disease; ATD, autoimmune thyroid disease; VIT, vitiligo; AST, asthma; ALL, allergies; SCZ, schizophrenia; T2D, type 2 diabetes; CAD, coronary artery disease.

Next, we applied mediated expression score regression (MESC)<sup>16</sup> to investigate the causal relationship between caQTLs in LCLs and IMD associations (Supplementary Figure 1A). Compared to S-LDSC analysis that tests for heritability enrichment of SNPs with some functional annotation (*e.g.*, accessible regions), MESC analysis specifically estimates the heritability that is mediated (*i.e.*,  $h^2_{med}$ ) by the SNPs' *cis*-effects on a molecular phenotype (*e.g.*, caQTLs). We estimated that caQTLs in LCLs mediate 16.3–42.7 % of autoimmune disease heritability and 8.5–9.4 % of allergic disease heritability (Figure 1B). For non-immune diseases, the estimates were lower and not significant ( $p > 0.003125$  (Bonferroni-corrected), MESC). Interestingly, schizophrenia showed a nominally significant proportion of caQTL-mediated heritability in LCLs ( $p < 0.05$ , MESC), consistent with the hypothesis that B cells may play some role in schizophrenia pathogenesis<sup>50,51</sup>. Our results indicate that LCLs are a valid cell type in which to search for caQTLs that mediate genetic risk for 7 IMDs — CD, IBD, MS, PBC, RA, SLE, and UC — but not for allergic diseases. In subsequent analyses, we focused on 7 IMDs — CD, IBD, MS, PBC, RA, SLE, and UC — that showed significant caQTL-mediated heritability ( $p < 0.003125$  (Bonferroni-corrected), MESC).

## Regions with higher levels of accessibility and active histone marks explain most of caQTL-mediated heritability

To understand which features characterize accessible regions that mediate IMD heritability, we estimated  $h^2_{med}$  enrichment (proportion of  $h^2_{med}$  / proportion of peaks)<sup>16</sup> in specific sets of accessible regions. We found that peaks with a larger number of nonredundant sequencing reads (*i.e.*, “stronger” peaks) in LCLs showed stronger  $h^2_{med}$  enrichment (Figure 1C) and thus likely affect IMD-relevant gene expression more than “weaker” peaks do. This observation is consistent with the “Activity-by-Contact” model<sup>9</sup>, in which peaks with greater chromatin accessibility and H3K27ac ChIP-seq signal are predicted to have proportional effects on target gene expression.

Next, we considered peaks with the active histone marks H3K27ac, H3K4me1, or H3K4me3<sup>28,52</sup>. Consistent with prior observations that putative cell-type-specific regulatory elements marked with H3K27ac and H3K4me1 are enriched for relevant disease associations<sup>34,35</sup>, we found that caQTLs with H3K27ac and H3K4me1 marks in LCLs were significantly enriched for mediated IMD heritability ( $q$ -value  $< 0.05$ , MESC; Figure 1D). Strikingly, both peak sets explained almost all of caQTL-mediated IMD heritability (Supplementary Figure 2). Peaks with H3K4me3 marks, representative of promoters<sup>53</sup>, also showed significant  $h^2_{med}$  enrichment for most IMDs ( $q$ -value  $< 0.05$ , MESC). Peaks with promoter-like signatures (*i.e.*, H3K27ac and H3K4me3)<sup>54</sup> and those with enhancer-like signatures (*i.e.*, H3K27ac, but no H3K4me3)<sup>54</sup> were also enriched for all IMD heritability ( $q$ -value  $< 0.05$ , MESC). Conversely, peaks without any of the three active histone marks were completely depleted of caQTL-mediated IMD heritability (Supplementary Figure 2). These ‘ATAC only’ peaks were shorter, weaker, and further away from the TSS compared to peaks with active histone marks (Supplementary Figure 3). Altogether, these results indicate that peaks characterized as putatively active regulatory elements explain nearly all of caQTL-mediated IMD heritability.

## caQTLs share IMD heritability with eQTLs and explain more of IMD heritability than do eQTLs

The model that gene regulatory activity explains a significant fraction of noncoding genetic associations to IMDs is supported by our findings that caQTLs mediate a significant proportion of IMD heritability and that those with active histone marks show strong  $h^2_{med}$  enrichment. This is in contrast with relatively low average  $h^2_{med} / h^2_{SNP}$  estimates ( $\sim 11\%$ ) previously having been observed for eQTLs across 48 human tissues in GTEx and various human traits<sup>55</sup>. To directly compare the proportion of IMD heritability mediated by caQTLs and eQTLs in the same cell type (*i.e.*, LCLs), we additionally applied MESC to gene expression data from LCLs (*i.e.*, Geuvadis data, Supplementary Figure 1B)<sup>56</sup>.

Across the 7 autoimmune diseases, the estimated proportion of heritability mediated by eQTLs ( $h^2_{\text{med; eQTL}} / h^2_{\text{SNP}}$ ) ranged from 9% to 22% (**Figure 2A**). For all 7 diseases, we estimated that eQTLs mediated less heritability than did caQTLs, even though the caQTLs' smaller sample size would potentially bias the estimates toward zero<sup>16</sup>. A possible explanation is that some IMD-associated regulatory variants may show detectable effects on chromatin accessibility, but not on gene expression, in LCLs at the current sample size ( $n = 373$ ); such loci may account for the missing regulation.

We anticipated that IMD-associated variants that affect gene expression in *cis* do so by modulating regulatory element's activity. Therefore, we investigated whether eQTL-mediated IMD heritability is shared by caQTL-mediated signals (**Figure 2B**). We performed MESC on both caQTLs and eQTLs together to estimate the amount of IMD heritability mediated by both collectively (**Methods**). For the 7 IMDs, the combined  $h^2_{\text{med; caQTL} \cup \text{eQTL}} / h^2_{\text{SNP}}$  was only slightly higher (2.2–9.0 %) than the estimates for just caQTLs ( $h^2_{\text{med; just caQTL}} / h^2_{\text{SNP}}$ ; **Figure 2A**), suggesting that approximately 56–82% of eQTL-mediated heritability is shared with caQTL-mediated heritability (i.e.,  $h^2_{\text{med; caQTL} \cap \text{eQTL}} / h^2_{\text{med; eQTL}}$ ; **Supplementary Figure 4**). These estimates are consistent with substantial sharing of caQTL- and eQTL-mediated IMD heritability. Nevertheless, 9–27% of IMD heritability is explained just by caQTLs, while only 2–9% of IMD heritability is explained just by eQTLs.

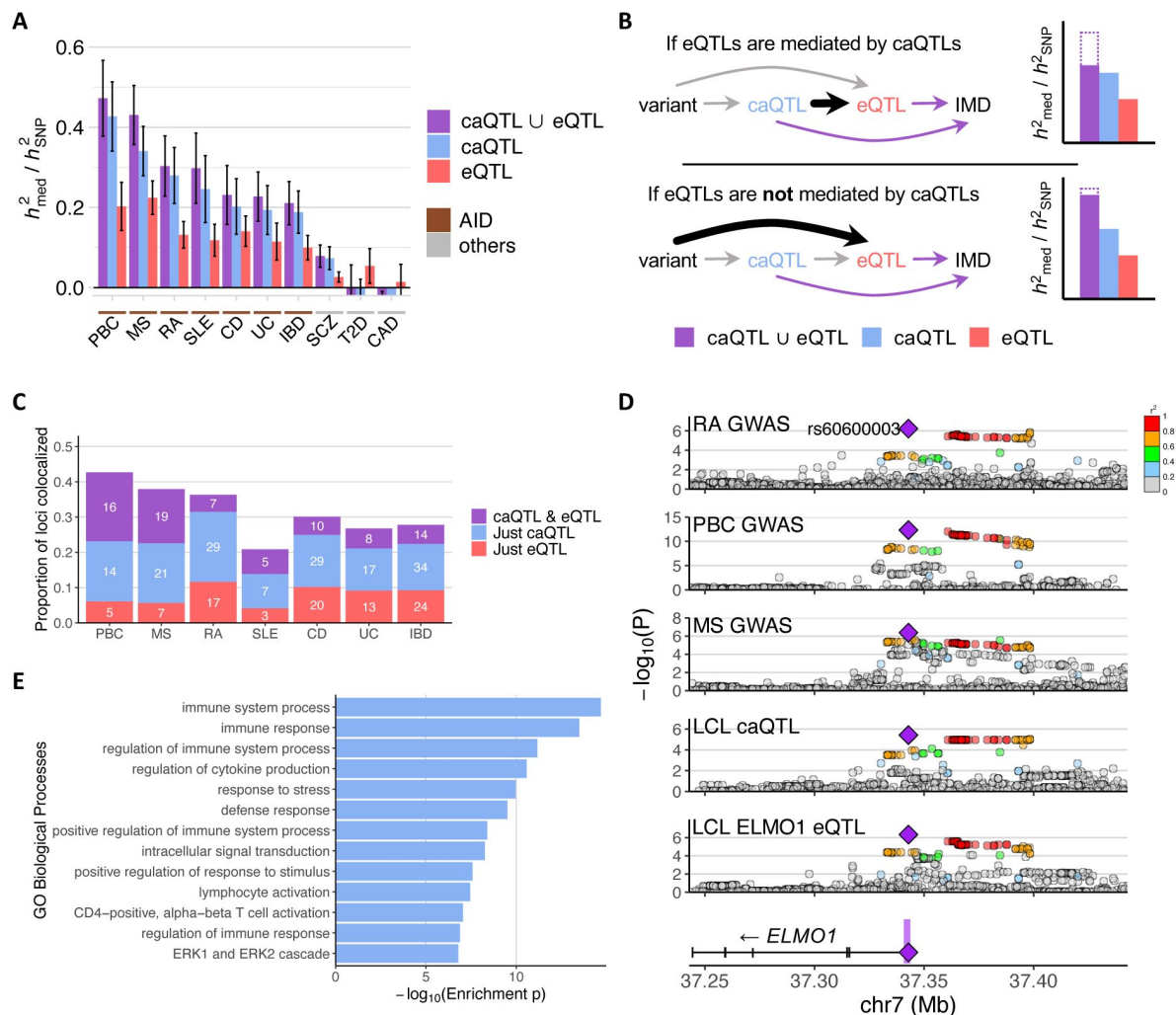
## Many IMD-associated loci show colocalization with caQTLs but not with eQTLs

We applied colocalization analysis<sup>37</sup> to identify IMD-associated loci that share genetic signals with caQTLs or eQTLs in LCLs. We selected candidate loci of 200-kb windows for each IMD with the following conditions: (1) lead IMD association at  $p < 10^{-6}$ , (2) lead caQTL or eQTL association at  $p < 10^{-4}$ , and (3) at least one variant simultaneously showed caQTL or eQTL  $\chi^2$  statistics greater than  $0.8 \times$  lead  $\chi^2$  statistics for the caQTL or eQTL, respectively, and IMD association  $\chi^2$  statistics greater than  $0.8 \times \chi^2$  statistics for the IMD lead variant in the locus. We applied gwas-pw<sup>37</sup> and considered loci with posterior probability of colocalization (PPA3)  $> 0.98$  to be colocalized<sup>57</sup>. Some loci colocalized with only either a caQTL or an eQTL, while others colocalized with both (**Figure 2C** and **Supplementary Tables 1–2**).

We investigated which proteins might be interacting with the colocalized caQTL peaks using Cistrome<sup>58</sup>. We tested for overlap of the colocalized caQTL peak regions with ChIP-seq peaks detecting diverse proteins in immune-related cells (**Supplementary Table 3**). Consistent with the enriched mediated heritability in accessible regions with active histone marks (**Figure 1D** and **Supplementary Figure 1**), proteins that are most often detected at the colocalized accessible regions are those related to RNA transcription (POL2RA and MED1), chromatin remodeling (EP300, BRD4, SMARCA4, and MTA2) or immune cell transcription factors (IKZF1, RUNX3, SPI1, RELA, RUNX1, and EBF1) (**Supplementary Figure 5**). Interestingly, TRIM28, which functions as a repressor, was one of the most overlapping protein factors.

To confirm that the colocalized genes are relevant to IMD, we tested for their enrichment of Gene Ontology (GO) annotation terms for specific biological processes<sup>59</sup>. Considering all genes within 500 kb of the IMD GWAS lead variants at colocalized loci as background, the genes that showed eQTL colocalization for any IMD were enriched for various immune responses and signaling processes, such as “positive regulation of immune system process” and “regulation of lymphocyte activation” (**Supplementary Figure 6**), indicating that the colocalized genes in LCLs are involved in immune function. For example, *IL6R* and *IL12A* encode direct or indirect targets of approved drugs – Tocilizumab and Ustekinumab – for autoimmune diseases like RA<sup>60</sup> and CD<sup>61</sup>. These two genes showed colocalization with both caQTLs and eQTLs in CD and PBC GWAS, respectively (**Supplementary Figure 7A–B**). Increased *IL6R* expression was associated with higher risk for CD, and increased *IL12A* expression was associated with lower risk for PBC and SLE. The former





**Figure 2.**

### IMD heritability mediated by caQTLs and eQTLs.

(A)  $h^2_{med} / h^2_{SNP}$  estimates of various IMDs for caQTLs, eQTLs, and their union. AID: autoimmune disease. Disease abbreviations along the x-axis are as in [Figure 1](#). (B) Schema of the potential causal relationships between genetic variants, caQTLs, eQTLs, and IMD risk. The two diagrams depict possible  $h^2_{med} / h^2_{SNP}$  trends depending on the causal relationship between caQTLs and eQTLs. (C) Number of IMD-associated loci colocalized with caQTLs or eQTLs. The proportion is out of the total number of IMD-associated ( $p < 10^{-6}$ ) loci. (D) *ELMO1* locus plot showing association to RA, PBC, MS, chromatin accessibility, and *ELMO1* expression in LCLs. Purple shading in the gene plot at the bottom indicates the caQTL peak, and the purple diamond is the lead variant (rs60600003) that is within that peak. The other variants are colored by the degree of linkage disequilibrium (LD) with the annotated variant. (E) Enrichment of the Biological Process Gene Ontology (GO) terms of genes in proximity to IMD-colocalized caQTLs without eQTL colocalization.

observation is in line with Tocilizumab, a monoclonal antibody to IL-6 receptor, showing efficacy in CD patients<sup>60</sup>, although it is not pursued for approval because of potential side effects<sup>62</sup>. Interestingly, *IL6R* and *IL12A* eQTLs did not colocalize with the association signals of RA and CD, respectively, which are the diseases for which these drugs are approved (Supplementary Figure 7C-D). Moreover, *ELMO1*, which previously had not been associated with autoimmune diseases, showed eQTL colocalization with RA, PBC, and MS association signals (Figure 2D). In all three, decreased *ELMO1* expression was associated with increased disease risk. In mice, *Elmo1* was required for polarization and migration of B and T lymphocytes<sup>63</sup>.

Across the IMDs, there were many loci that colocalized with a caQTL but not with an eQTL (Figure 2C). These “caQTL only” loci showed enrichment for immune response genes in *cis* compared to all accessible regions in LCLs<sup>64,65</sup>, even though the colocalized eQTLs were enriched for immune response genes as well (Figure 2E and Supplementary Figure 6), indicating that IMD-relevant genes without eQTL colocalization in Geuvadis LCL data<sup>56</sup> are likely found in these loci.

## Distance to transcription start sites affect eQTLs but not caQTLs

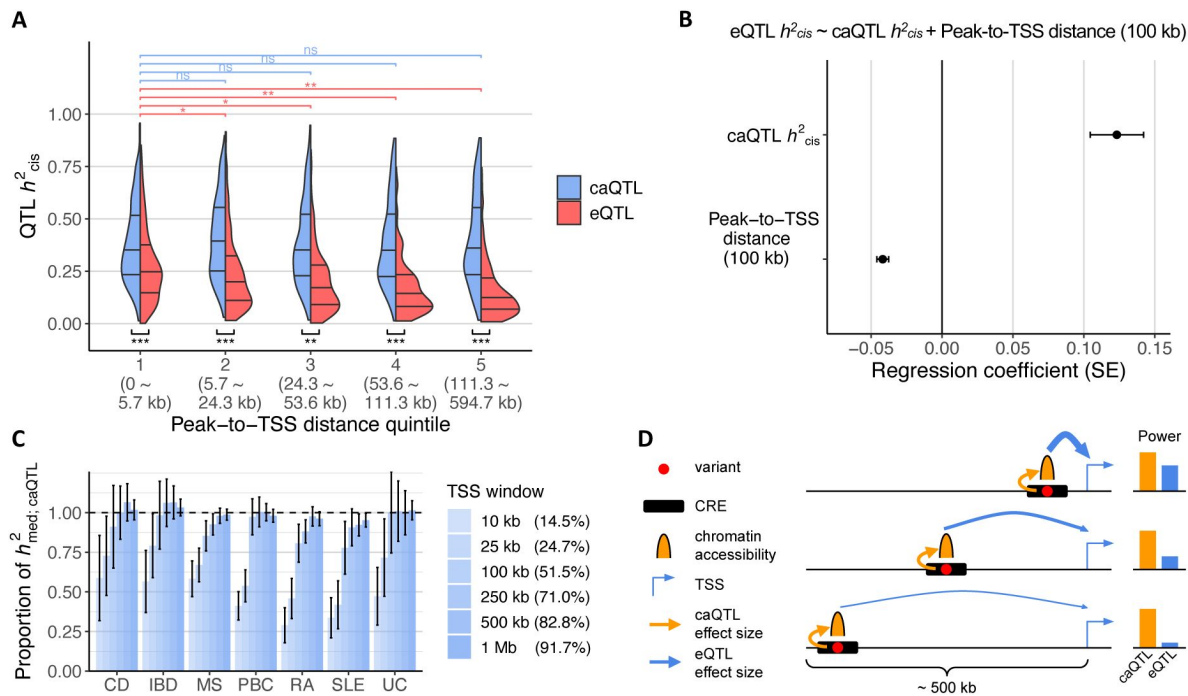
Why might there be loci with caQTL colocalization only, despite the caQTL data having fewer samples than the eQTL data (100 vs. 373)? Limited statistical power can prevent some eQTLs from being detected and showing significant colocalization<sup>19</sup>. As hypothesized by Mostafavi and colleagues, disease-relevant eQTLs may be weaker and more distal<sup>18</sup>. To understand the extent to which this effect may result in many loci showing colocalization only with caQTLs, we compared the *cis*-heritability ( $h^2_{cis}$ ) of caQTLs and eQTLs depending on the distance from the ATAC peak to the transcription start site (TSS) of the gene (*i.e.*, peak-to-TSS distance). We considered all caQTLs and eQTLs regardless of disease association.

We identified pairs of caQTLs and eQTLs that colocalized with each other<sup>37</sup>, which implies that the regulatory variant modulating chromatin accessibility also affects gene expression. Then, the distance between the ATAC peak and the TSS of the eQTL gene (*i.e.*, eGene) is the distance between a regulatory element and its target gene's TSS. We stratified the pairs into peak-to-TSS distance quintiles and compared the eQTL  $h^2_{cis}$  distribution of the first quintile (*i.e.*, closest pairs) with that of the later quintiles. We observed that eQTL  $h^2_{cis}$  distribution decreased with increasing distance of the paired ATAC peaks from the TSS ( $p = 1.0 \times 10^{-4}$ ,  $2.1 \times 10^{-10}$ ,  $1.6 \times 10^{-15}$ , and  $1.7 \times 10^{-20}$ , respectively, one-sided Wilcoxon rank-sum test; Figure 3A), consistent with the negative relationship between promoter-enhancer genomic distance and impact on gene expression<sup>66,67</sup>. This result also explains why discovered eQTLs are concentrated near the promoter, where the variants are more likely to show stronger effects<sup>68</sup>. In contrast, caQTL  $h^2_{cis}$  distribution was similar across peak-to-TSS distances ( $p > 0.05$ , one-sided Wilcoxon rank-sum test; Figure 3A). For all distance quintiles, caQTL  $h^2_{cis}$  was significantly higher than that of the paired eQTLs ( $p = 4.0 \times 10^{-23}$ ,  $1.3 \times 10^{-24}$ ,  $1.5 \times 10^{-19}$ ,  $1.1 \times 10^{-28}$ , and  $2.4 \times 10^{-49}$ , respectively, one-sided paired Wilcoxon rank-sum test; Figure 3A), and the contrast between them was greater at more distant quintiles, suggesting that the statistical power to detect and colocalize eQTLs is increasingly lower than that for caQTLs for regulatory effects far from the TSS.

Overall, caQTL  $h^2_{cis}$  had a significant positive effect ( $p = 6.6 \times 10^{-11}$ , linear regression) and peak-to-TSS distance had a negative effect ( $p = 2.4 \times 10^{-23}$ , linear regression) on eQTL  $h^2_{cis}$  (Figure 3B). Thus, for regulatory variants that showed both caQTL and eQTL signals, those with larger effects on chromatin accessibility tended to exhibit larger effects on gene expression, but their eQTL effects diminished with increasing distance from TSSs.

Next, we investigated how caQTL-mediated IMD heritability is distributed with respect to TSS. If caQTLs beyond the typical *cis*-eQTL window of 1 megabase (Mb) around the genes' TSS explain some proportion of IMD heritability, then *cis*-eQTL analyses might require a wider window to detect disease-associated eQTLs. Across the 7 diseases, caQTLs within 500 kb of the TSS of





**Figure 3.**

### Effect of peak-to-TSS distance on *cis*-heritability of caQTLs and eQTLs and IMD heritability.

(A) Distribution of *cis*-heritability ( $h^2_{cis}$ ) of caQTLs and eQTLs by peak-to-TSS distance quintiles. The ranges of peak-to-TSS distance are shown in parentheses. The comparisons shown on the top (in respective colors) are between the nearest and each subsequent quintile of the respective QTL  $h^2_{cis}$  distribution (i.e., one-sided Wilcoxon rank-sum test). The comparisons shown on the bottom (in black) are between caQTL and eQTL  $h^2_{cis}$  distribution (one-sided paired Wilcoxon rank-sum test). \*:  $p < 10^{-4}$ , \*\*:  $p < 10^{-10}$ , \*\*\*:  $p < 10^{-20}$ , and ns:  $p > 0.05$ . (B) Regression estimates and their standard errors of the linear regression model testing the effects of caQTL  $h^2_{cis}$  and peak-to-TSS distance on eQTL  $h^2_{cis}$ . Peak-to-TSS distance was expressed in units of 100 kb to neatly visualize the effect size estimates. SE: standard error. (C) Proportion of caQTL-mediated IMD heritability explained by ATAC peaks within various TSS windows. Percentage for each TSS window denotes the proportion of ATAC peaks in that window. Disease abbreviations along the x-axis are as in Figure 1. (D) A model of the relationship between peak-to-TSS distance and power to detect a corresponding caQTL or eQTL. The thickness of the arrows indicates the variant effect size on chromatin accessibility (yellow) or gene expression (blue). TSS, transcription start site; CRE, *cis*-regulatory element.

expressed genes explained almost all of the caQTL-mediated IMD heritability (92–100%; **Figure 3C**), indicating that regulatory variants are most likely within 500 kb of the target gene's TSS and supporting the use of a 1 Mb window for *cis*-eQTL analyses. Depending on the disease, 41–66% of the caQTL-mediated IMD heritability was detected in distal peaks further than 10 kb from the TSS of expressed genes, further supporting the analysis of regulatory variants beyond promoter regions.

In sum, the power to detect eQTLs diminishes with increasing distance of the variant from the TSS, but the power to detect caQTLs is largely invariant regardless of peak-to-TSS distance (**Figure 3D**). Since  $h^2_{\text{med; caQTL}}$  are distributed mostly within 500 kb of genes' TSS, the IMD loci colocalizing only with caQTLs could still be weak, undetected eQTLs. Under this model, we predicted that increasing the power to detect eQTLs in LCLs, such as increasing sample size, may lead to further eQTL colocalizations in loci in which we observed only caQTL colocalization.

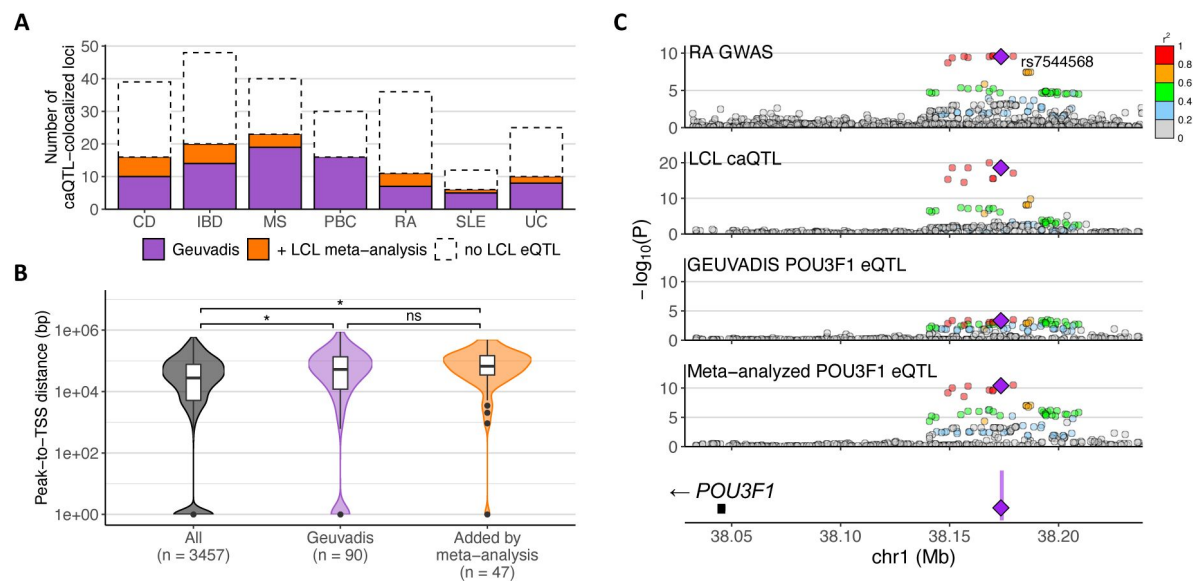
## Increasing the sample size reveals some eQTL colocalization

For genetic association studies, increasing the sample size is a way to increase statistical power. Therefore, we meta-analyzed four LCL eQTL summary statistics<sup>55,56,69,70</sup>, leading to a total sample size of 1,128 individuals. We performed colocalization analysis using the meta-analyzed summary statistics to evaluate whether effectively increasing the sample sizes would uncover more disease-associated eQTLs in LCLs, especially in loci where a caQTL already showed IMD colocalization. Up to 6 additional loci showing eQTL colocalization were thus detected for each IMD (**Figure 4A** and **Supplementary Table 4**). For example, *CIITA* is the class II major histocompatibility complex transactivator, which causes severe immunodeficiency if dysfunctional<sup>71</sup>. The *CIITA* locus is associated with IBD right below the genome-wide significance level (rs10445003,  $p = 7.5 \times 10^{-8}$ ), and it colocalized with a caQTL signal, but initially not with any eQTL (**Supplementary Figure 8A**). However, the meta-analyzed statistics showed a stronger association to *CIITA* expression ( $p = 6.7 \times 10^{-8}$ ) than without meta-analysis ( $p = 5.4 \times 10^{-4}$ ) and exhibited a significant colocalization. Interestingly, two of the causal CD genes that previously lacked colocalized eQTLs<sup>15</sup>, *CARD9* and *ATG16L1*, showed significant colocalization in the meta-analyzed LCL eQTL data (**Supplementary Figure 8B-C**).

We hypothesized that increased sample size would improve the power to detect weaker and distal eQTL colocalization. Comparison of the accessibility peak's distance to the paired eQTL gene's (eGene) TSS showed that the newly detected eQTLs tended to be more distal ( $p = 0.06$ , one-sided Wilcoxon rank-sum test; **Figure 4B**). However, compared to the distribution of the peak-to-TSS distance for all caQTL-eQTL pairs showing colocalization, IMD-associated loci that showed caQTL and eQTL colocalization had greater peak-to-TSS distance on average ( $p = 0.002$  for Geuvadis and  $p = 5.9 \times 10^{-6}$  for the meta-analyzed data; **Figure 4B**). For example, a RA-associated locus near *POU3F1*, a neuronal transcription factor that is also induced by interferon<sup>72</sup>, colocalized with a distal eQTL located about 126 kb upstream of its promoter, after meta-analysis strengthened the eQTL association ( $p < 10^{-10}$ ; **Figure 4C**). These results suggest that additional IMD-associated loci with distal, weaker eQTLs in LCLs might be found if eQTL data were generated for a larger number of individuals.

## IMD loci that colocalized with caQTLs but not eQTLs showed lower levels of active histone marks

Despite uncovering more eQTL colocalizations through meta-analysis, more than 40% of the caQTL-colocalized loci nevertheless showed no colocalization with an eQTL in LCLs (**Figure 4B**). We investigated whether these “caQTL only” peaks might be inactive regulatory elements. We quantified the active histone mark levels for H3K27ac, H3K4me1, and H3K4me3 at colocalized caQTL peaks in LCLs and then compared their levels between the “caQTL and eQTL” and “caQTL only” loci. On average, H3K27ac marks were stronger at “caQTL and eQTL” peaks, supporting that the corresponding regulatory elements might be more active (**Supplementary Figure 9**).



**Figure 4.**

#### Additional colocalization of caQTL-colocalized IMD loci with meta-analyzed LCL eQTL data.

(A) Number of caQTL-colocalized IMD loci that showed eQTL colocalization in LCLs. Disease abbreviations along the x-axis are as in [Figure 1](#). (B) Distribution of peak-to-TSS distance of all caQTL-eQTL pairs and of those colocalized with IMD association. The number of loci in each category is shown in parentheses. \*:  $p < 0.01$ , ns:  $p > 0.05$ . (C) *POU3F1* eQTL that became significantly colocalized with RA association by meta-analyzing LCL eQTL data. Purple shading in the gene plot at the bottom indicates the caQTL peak, and the purple diamond is the lead variant (rs60600003) that is within that peak. The other variants are colored by the degree of linkage disequilibrium (LD) with the annotated variant.

H3K4me3 marks were detected more often at “caQTL and eQTL” peaks, leading to stronger average signal. In contrast, H3K4me1 levels were highly similar between the two sets of peaks. Although “caQTL only” peaks generally showed lower levels of active histone marks, several individual “caQTL only” peaks showed comparable levels. These peaks could be inactive *cis*-regulatory elements in LCLs that affect gene expression in a different cellular context. Therefore, we next examined whether those caQTLs appear as eQTLs in other immune cell types.

## Various immune cell types exhibit eQTL colocalization where LCLs did not

We downloaded eQTL summary statistics generated from 26 naïve and stimulated immune cell types<sup>22,23,73–75</sup> to search for eQTLs that may correspond to the remaining, IMD-colocalized caQTLs (Supplementary Table 5). The profiled cell types range from B cells and monocytes to subtypes of T cells, as well as stimulated T cells and macrophages. We tested for colocalization of these eQTLs to IMD associations.

25–42% of the caQTL-colocalized IMD loci that were missing eQTL colocalizations in LCLs showed eQTL colocalizations in at least one immune cell type (Figure 5A and Supplementary Table 6). The overlap of LCL caQTLs with non-LCL immune cell eQTLs was greater than expected by chance for 5 of the 7 IMDs ( $p < 0.00714$  (Bonferroni-corrected), Fisher’s exact test, for CD, IBD, PBC, RA, and UC; Supplementary Table 7), suggesting that the caQTLs found in LCLs may also show regulatory function in those immune cell types. Comparing across the datasets, we found that the number of loci with eQTL colocalization varied depending on the cell type, but that the effect of the sample size was more profound ( $r^2 = 0.60$ – $0.79$ ; Figure 5B and Supplementary Table 8). We meta-analyzed eQTL data of 3 immune cell types with multiple sources – naïve CD4<sup>+</sup> T cell, monocyte, and memory regulatory T cell (Treg) – and this also increased the number of loci with significant eQTL colocalization (Supplementary Table 9). Altogether, these results suggest that although generating eQTL data in more cell types and cell states uncovers context-specific eQTLs, increasing the sample size should also be a priority to ensure sufficient statistical power.

We investigated the potential reasons why IMD loci that colocalized with caQTLs in LCLs showed eQTLs not in LCLs but in other immune cells. First, LCL caQTLs may correspond to gene regulatory elements that exert their effects on gene expression in a different cellular context. For instance, monocyte H3K27ac levels in the “caQTL only” loci where monocyte eQTLs colocalized were higher than those with no monocyte eQTLs (Supplementary Figure 10). Second, some examples, such as for *TNFSF15*, were due to cell-type specific gene expression (Figure 6): despite a significant colocalization of IBD with a caQTL in LCLs in the *TNFSF15* locus, disease-associated eQTL signal was detected only in monocytes (Figure 6A). Tumor necrosis factor-like cytokine 1A (TL1A), the protein encoded by *TNFSF15*, is secreted by monocytes and many other cells to activate helper T cells, regulatory T cells, and B cells<sup>76</sup>. *TNFSF15* expression was low in LCLs (mean transcript per million [TPM] = 0.30), but higher in monocytes (mean TPM = 2.23). Of the genes that showed exclusively monocyte eQTL colocalization, those with low expression (mean TPM < 1) in LCLs generally showed higher expression in monocytes (Figure 6B–C). However, low expression in LCLs was likely not the explanation for most cases of “monocyte only” eQTLs (blue points in Figure 6B) because most “monocyte only” eQTL genes were expressed at level higher than 1 TPM in LCLs.

Overall, expanding the eQTL search to various immune cell types increased the number of eQTL-colocalized loci among those that previously colocalized only with caQTLs (Figure 5A). On average, approximately 75% of the caQTL-colocalized loci ultimately showed eQTL colocalization. These results highlight the utility of eQTL data across a range of immune cell types for discovery of IMD-associated eQTLs.

Figure 5.

Added utility of various immune cell eQTL data.

(A) Number of loci that additionally colocated with eQTLs by LCL meta-analysis (orange) and immune cell data (cyan) compared to the original analysis with Geuvadis LCL eQTL data (purple). The height of the bar is the proportion of loci with eQTL colocalization out of the total IMD loci with caQTL colocalization in the earlier analysis. Disease abbreviations along the x-axis are as in Figure 1. (B) Relationship between the number of MS GWAS loci with eQTL colocalization and sample size for each eQTL dataset. Meta-analyzed eQTL data are labeled with their cell types. NK cell, natural killer cell; Treg, regulatory T cell.

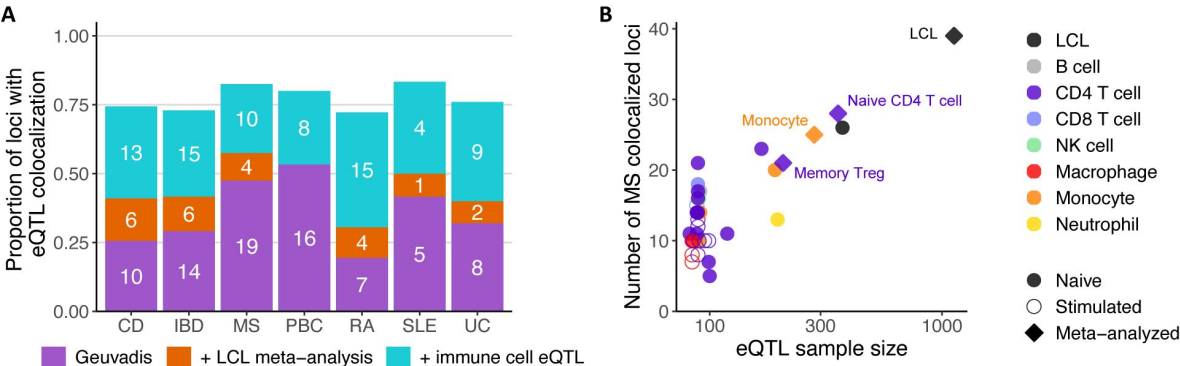
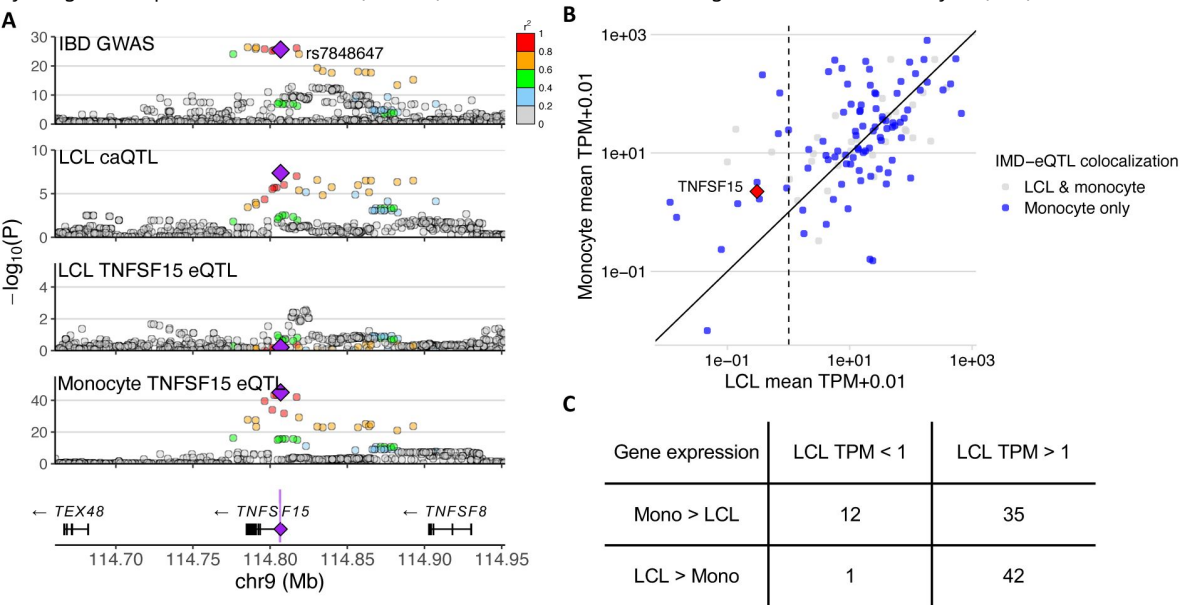


Figure 6.

IMD loci that colocated with an eQTL in monocytes, but not in LCLs, even though they colocated with caQTLs in LCLs.

(A) *TNFSF15* locus plot showing genetic association to IBD, chromatin accessibility in LCLs, and *TNFSF15* expression in LCLs and monocytes. Purple shading in the gene plot at the bottom indicates the caQTL peak, and the purple diamond shows a strongly associated variant (rs7848647) that is within that peak. The other variants are colored by the degree of LD with the annotated variant. (B) Expression levels of genes in LCLs and monocytes colored according to their eQTL colocalization outcome. TPM: transcripts per million. (C) Number of genes with eQTL colocalization in monocytes, but not in LCLs, separated by the gene's expression level in LCLs (column) and whether it is lower or higher than that in monocytes (row).





## Discussion

A lack of the link between many noncoding GWAS loci to the associated variants' gene regulatory effects have posed challenges in understanding their genetic mechanism<sup>15</sup>. Various hypotheses have been presented, including disease genes showing more complex gene regulation<sup>77</sup>, context specificity of gene regulation<sup>17</sup>, and a combination of both, due to selective constraints against damaging eQTLs<sup>18</sup>. A better evaluation of the potential reasons for the missing regulation will guide future data generation projects to elucidate disease-associated loci. To determine why some loci might lack colocated eQTLs, we focused on chromatin accessibility, which is a molecular phenotype affected by regulatory variants more directly than are eQTLs. We approached this question with mediated heritability analysis<sup>16</sup> and colocalization analysis<sup>36,37</sup>.

We found that caQTLs in LCLs mediate a significant proportion of heritability for many autoimmune diseases. In contrast, LCLs did not appear to be an effective cell type to model gene regulatory effects in allergic diseases. The  $h^2_{med} / h^2_{SNP}$  estimates for caQTLs were higher than those of eQTLs in most autoimmune diseases, even though the smaller sample size of caQTL data (i.e., 100 vs. 373) could bias caQTLs' estimates toward zero<sup>16</sup>. We also showed that disease-associated chromatin accessibility effects often share the genetic signal with gene expression effects but that there are also many loci without an eQTL detected in LCLs.

By focusing on disease-associated caQTL that lacked significant eQTLs, we explored how additional colocated eQTL effects could be uncovered. First, increasing the sample sizes for the eQTL statistics via meta-analysis demonstrated that more eQTL colocated effects can be detected with increased statistical power and robustness<sup>19</sup>. These results are consistent with the hypothesis that disease-associated eQTLs are typically weaker and distal due to negative selection against large expression changes for causal genes<sup>18</sup>. Second, many caQTLs in LCLs without eQTLs in LCLs showed eQTL colocalization in other immune cell types. Context specificity of eQTLs has been widely considered to be the primary explanation for the difficulty of pinpointing disease-associated eQTLs<sup>17,22,23,74</sup>. Our observation that many IMD-colocalized caQTLs in LCLs show eQTLs in other immune cell types suggests that caQTL effects may be shared across cell types whereas eQTL effects are more context-specific<sup>23</sup>. If a shared set of transcription factors are expressed in similar yet distinct cell types, like the immune cells, genetic variants affecting their DNA binding would affect chromatin accessibility similarly. On the other hand, regulation of gene expression is a result of multiple regulatory elements, each binding multiple transcription factors<sup>78</sup>, so the measured effect of a regulatory variant on gene expression likely depends much more on the cellular context. In such cases, eQTL data for the specific cellular contexts need to be generated in future studies to uncover genetic signal shared with a complex trait or disease. All in all, our results suggest that both increasing the sample size and generating gene expression data from more relevant cellular contexts would be useful strategies for discovering more disease-associated eQTLs.

Finally, we demonstrated that caQTLs can reveal the regulatory variant effect of disease-associated variants that may have been difficult to detect with eQTLs, particularly in TSS-distal regions. Although caQTLs cannot directly identify the target gene or the causal cellular context, we anticipate that integrated analyses can improve the power to detect weaker eQTL signals, as multi-trait GWAS analyses have shown<sup>79</sup>. Moreover, integrating multiple molecular QTL data, like transcription factor binding QTLs and histone mark QTLs, may highlight the regulatory elements associated with the GWAS phenotype, which may ultimately contribute to identifying the causal gene<sup>33</sup>. Therefore, we anticipate that generation of data across the various molecular phenotypes upstream of gene expression for QTL analyses will be informative.

## Limitations of the study

The  $h^2_{med} / h^2_{SNP}$  estimates can be biased because of insufficient sample size of the QTL data. Thus, the proportion of mediated IMD heritability could change based on the specific caQTL and eQTL data. Colocalization analysis tests whether the QTL and GWAS data likely share genetic signal, and such shared signal could arise from either causal mediation or pleiotropy. Therefore, further experiments are needed to establish causality of colocalized eQTL genes. Lastly, we hypothesized that caQTL effects are often shared across cell types, but we did not have access to relevant data to test this hypothesis.

## Methods

### ATAC-seq data processing and peak calling

We downloaded LCL ATAC-seq data of British (GBR) samples ( $n = 100$ ) from European Nucleotide Archive (ENA) under accession ERP110508<sup>30</sup>. The available files were cram alignment files mapped to the b37 reference genome, so we extracted unique read pairs using SAMtools<sup>80</sup> and bamtofastq command from bedtools<sup>81</sup>. The reads were paired-end and each 75 base pairs (bp) long. The data contained reads with Nextera transposase adapters, so we removed the adapter sequences and bases of poor quality at the 3' end using cutadapt. Trimmed reads with both pairs shorter than 20 bp were discarded. The command was “cutadapt -a file:\${forward} -A file:\${reverse} -e 0.25 -j 2 -q 15 --pair-filter=both -m 20”. ( \${forward} and \${reverse} files contain the forward and reverse Nextera transposase adapters). We mapped the reads to the GRCh38 reference genome using Bowtie 2<sup>82</sup> with the “GRCh38\_noalt\_decoy\_as” index provided on the tool's website. The command was “bowtie2 --very-sensitive --no-mixed --no-discordant -I 20 -X 2000”. We kept only read alignments with mapping quality greater than 1. We also removed reads aligning to the mitochondrial genome, those overlapping ENCODE exclusion regions (file ID: ENCF356LFX)<sup>83</sup> and potential PCR duplicates using scripts from WASP<sup>84</sup>.

To represent peaks across the samples, we subsampled 3 million read pairs from each and pooled them. Then, we used MACS2<sup>85</sup> with the BAMPE option for peak calling. The command was “macs2 callpeak -f BAMPE -g hs -q 0.05”. We further used the ‘bdgcmp’ and ‘bdgpeakcall’ subcommand to find peaks that are at least 100 bp long (-l) and merge those that are less than 100 bp apart (-g). We also merged peaks similarly derived from individual samples using the ‘merge’ command in bedtools.

Furthermore, for the sake of comprehensiveness, we repeated these steps with LCL ATAC-seq data of Yoruban (YRI) samples<sup>86</sup> and merged the peaks with the earlier peak set derived from GBR samples. In total, there were 443,403 peaks genome-wide.

### RNA expression data preparation

We downloaded RNA expression level data of the LCL samples<sup>56</sup> from the Expression Atlas<sup>87</sup>. These data consisted of TPM values of genes as processed by the Expression Atlas. We retained TPM values of protein-coding and long noncoding RNAs in European samples for downstream analyses, including QTL analysis, mediated heritability analysis, and colocalization.

### LCL samples' genotype preparation

To utilize the genotype calls of the highest quality, we downloaded the high-coverage 1000 Genomes (1kG) Project data<sup>88</sup>. Of the European samples with ATAC-seq or RNA-seq data, 14 samples had genotypes derived from microarrays<sup>89</sup>, and the remaining samples had genotypes derived from the high-coverage whole-genome sequencing data. The samples with only

microarray-based genotypes that needed imputation are listed in [Supplementary Table 10](#). We lifted over the microarray data based on the hg19 reference genome to GRCh38 and filtered for variants present in the high-coverage 1kG data. We first imputed microarray genotype data using the TOPMed imputation server<sup>90</sup> and extracted SNPs with imputation  $R^2 \geq 0.5$  and imputed the rest of the variants, including short indels, using high-coverage data with Beagle5.2<sup>91,92</sup>. After keeping only variants with  $DR2 \geq 0.7$ , we merged the imputed genotypes with the high-coverage 1kG genotypes.

## Curation of IMD GWAS data

We downloaded GWAS summary statistics for 13 IMDs, including 11 autoimmune diseases (autoimmune thyroid disease (ATD)<sup>39</sup>, celiac disease (CEL)<sup>40</sup>, Crohn's disease (CD)<sup>41</sup>, inflammatory bowel disease (IBD)<sup>41</sup>, juvenile idiopathic arthritis (JIA)<sup>42</sup>, multiple sclerosis (MS)<sup>43</sup>, primary biliary cholangitis (PBC)<sup>39</sup>, rheumatoid arthritis (RA)<sup>44</sup>, systemic lupus erythematosus (SLE)<sup>45</sup>, ulcerative colitis (UC)<sup>41</sup>, and vitiligo (VIT)<sup>46</sup>) and 2 allergic diseases (allergy (ALL)<sup>47</sup> and asthma (AST)<sup>47</sup>). For each disease, we searched for more recent studies with larger sample sizes and prioritized those with genome-wide statistics, rather than those with only Immunochip variants<sup>93</sup>. To compare with non-immune diseases, we also downloaded summary statistics for 3 non-immune diseases (type 2 diabetes (T2D)<sup>48</sup>, coronary artery disease (CAD)<sup>49</sup> and schizophrenia (SCZ)<sup>50</sup>). In this study, we analyzed only those GWAS summary statistics derived from cohorts of individuals with European ancestries.

Since the LCL samples' genotypes are based on the GRCh38 reference genome, we lifted over any GWAS data based on b37 reference genome to the GRCh38 genomic coordinates. Briefly, we formatted the summary statistics as bed files and used the liftOver tool<sup>94</sup> to convert them to GRCh38 genomic coordinates. Then, to ensure that the reference alleles match the sequences of the GRCh38 reference genome, we used gwas2vcf tool<sup>95</sup>.

## Chromatin accessibility QTLs in lymphoblastoid cell lines

First, we quantified the chromatin accessibility levels at ATAC-seq peaks identified earlier. We counted the number of read fragments overlapping each peak using featureCounts<sup>96</sup>. For each sample, the read counts were normalized for library size using trimmed mean of M-values<sup>97</sup> so that the values are comparable across the samples. Then, the phenotype values were further normalized to follow a standard normal distribution across the samples, using quantile normalization. Peaks with counts per million (CPM)  $< 0.8$  or counts  $< 10$  for more than 20% of the samples were discarded.

Next, we performed a principal component analysis (PCA) on the phenotype matrix to derive potential latent covariates. We selected the number of principal components (PC) to incorporate in the regression model based on the Buja and Eyuboglu algorithm<sup>98</sup> that is implemented in PCAForQTL<sup>99</sup>. Ultimately, we accounted for sex, library size, 3 genotype PCs and 13 phenotype PCs in the QTL analysis. We performed genetic association tests on variants within 200 kilobases (kb) of the peak using tensorQTL<sup>100</sup>. We discarded variants with minor allele frequency less than 5%.

## IMD heritability enrichment in accessible regions of LCLs

We evaluated the relevance of accessible regions in LCLs to IMD heritability using stratified LD score regression (S-LDSC)<sup>38</sup>. We used the baselineLD v2.2 annotation in hg38 and the European LD reference from the 1000 Genomes Project (downloaded from the S-LDSC website, <https://alkesgroup.broadinstitute.org/LDSCORE/GRCh38/>). We used the set of filtered ATAC-seq peaks that we tested for QTL associations. We accounted for 16 diseases (13 IMDs and 3 non-IMDs) for the Bonferroni-corrected  $p$ -value threshold.

## Mediated heritability estimation of QTLs

We estimated the heritability mediated by QTLs ( $h^2_{\text{med}}$ ) using mediated expression score regression (MESR)<sup>16</sup>. We denote heritability of tested SNPs as  $h^2_{\text{SNP}}$ .

### caQTL-mediated heritability

We estimated the ‘expression scores’ for chromatin accessibility in LCLs using individual genotypes and phenotypes. We analyzed the same set of peaks and accounted for the same covariates as we did for the QTL analysis above. For mediated heritability estimation, we accounted for baselineLD v2.2 annotation in hg38 without the QTL annotations, as they could be redundant. The estimand of interest is the proportion of heritability mediated by caQTLs ( $h^2_{\text{med}} / h^2_{\text{SNP}}$ ).

To evaluate whether certain peak sets are enriched for mediated heritability, we utilized the gene set analysis functionality. For peak strength, we considered the 95% percentile CPM value of each peak and stratified the peaks into quintiles. For histone marks, we first generated histone ChIP-seq peaks using data from Delaneau *et al.*<sup>52</sup>. Similar to calling ATAC-seq peaks, we sampled 3 million reads per sample and merged them before applying MACS2<sup>85</sup>. We downloaded 3 control ChIP-seq data from ENCODE to use as input (File IDs: ENCFF066RCS, ENCFF159XTB, and ENCFF850RIE)<sup>83</sup>. Then, we curated sets of ATAC-seq peaks that overlapped H3K27ac, H3K4me1, and H3K4me3 ChIP-seq peaks by at least 1 bp. ATAC-seq peaks that overlapped H3K27ac but not H3K4me3 regions were labeled as ‘enhancer-like signature’, while those that overlapped H3K27ac and H3K4me3 regions were labeled as ‘promoter-like signature’. The estimand of interest is the proportion of mediated heritability explained by the peak set (peak set  $h^2_{\text{med}}$  / total  $h^2_{\text{med}}$ ).

### Comparison with eQTL-mediated heritability

First, we estimated  $h^2_{\text{med}} / h^2_{\text{SNP}}$  of eQTLs (*i.e.*,  $h^2_{\text{med; eQTL}} / h^2_{\text{SNP}}$ ) the same way as we did for that of caQTLs. Then, we also estimated  $h^2_{\text{med}} / h^2_{\text{SNP}}$  of caQTLs and eQTLs together (*i.e.*,  $h^2_{\text{med; caQTL} \cup \text{eQTL}}$ ) with MESR meta-analysis<sup>16</sup>. caQTLs and eQTLs were also stratified as separate sets to account for potential differences in the relationship of QTL *cis*-heritability and  $h^2_{\text{med}}$ . This meta-analyzed estimate is effectively the amount of heritability mediated by either caQTLs or eQTLs in LCLs. This estimate reveals the overall relationship between heritability mediated by caQTLs and eQTLs. For instance, the estimate of the heritability mediated exclusively by caQTLs would be  $h^2_{\text{med; just caQTL}} = h^2_{\text{med; caQTL} \cup \text{eQTL}} - h^2_{\text{med; eQTL}}$ . The estimate of mediated heritability shared by caQTLs and eQTLs is  $h^2_{\text{med; caQTL} \cap \text{eQTL}} = (h^2_{\text{med; caQTL}} + h^2_{\text{med; eQTL}}) - h^2_{\text{med; caQTL} \cup \text{eQTL}}$ .

## Colocalization analyses

### caQTL and eQTL colocalization with IMD GWAS

First, we selected candidate colocalization loci by filtering for overlapping ‘significant’ variants. The candidate loci met the following conditions: (1) lead IMD association at  $p < 10^{-6}$ , (2) lead caQTL or eQTL association at  $p < 10^{-4}$ , and (3) at least one variant simultaneously showed caQTL or eQTL  $\chi^2$  statistics greater than  $0.8 \times$  lead  $\chi^2$  statistics for the caQTL or eQTL and IMD association  $\chi^2$  statistics greater than  $0.8 \times \chi^2$  statistics for the IMD lead variant in the locus. Then, we applied gwas-pw<sup>37</sup> on the variants within 100 kb of the lead variant. We considered loci with posterior probability of colocalization (PP3) > 0.98 to be colocalized<sup>57</sup>.

## Colocalization of caQTL with eQTL

We performed a colocalization analysis of caQTLs and eQTLs to curate a set of loci where the same genetic signal likely explains both associations. The pairs of caQTLs and eQTLs reveal the distance between the regulatory element and the target gene's TSS. We selected candidate colocalization loci with: (1) IMD association at  $p < 10^{-6}$ , (2) QTL association at  $p < 10^{-4}$ , and (3) the two lead variants showed LD  $r^2 > 0.8$ . We applied gwas-pw<sup>37</sup> on the variants within 200 kb of the tested caQTL peak. We considered loci with posterior probability of colocalization (PP3)  $> 0.98$  to be colocalized<sup>57</sup>.

## Overlap of protein factor ChIP-seq and colocalized caQTL peaks

We searched for any protein factors detected at the colocalized caQTL peaks using the Cistrome database<sup>58</sup>, which we accessed on July 24, 2024. We considered only ChIP-seq data from immune cell types, progenitors and stem cells that can differentiate into immune cells. For each ChIP-seq peak, we searched for those that show overlap of more than 50% with one of 305 caQTL peaks that colocalized with IMD GWAS signals (Supplementary Table 1). Each protein factor detected in the caQTL peaks and the cell type used to generate the ChIP-seq data are listed in Supplementary Table 3.

## Enrichment of biological processes

To test whether colocalized genes are likely relevant to autoimmune diseases, we surveyed which Biological Process Gene Ontology (GO) terms were overrepresented compared to all the genes within 500 kb of each IMD association signal. Enrichment of biological processes was evaluated using Protein Analysis Through Evolutionary Relationships (PANTHER)<sup>101</sup>. The foreground list comprised all of the genes whose eQTL signal colocalized with one of the 7 autoimmune diseases (CD, IBD, MS, PBC, RA, SLE, and UC). The background list of genes was all of the genes within 500 kb of each IMD lead variant for which we observed colocalization. Moreover, we tested whether colocalized caQTLs without eQTLs are closer to genes related to immune processes than expected by chance. For this, we used Genomic Regions Enrichment of Annotations Tool (GREAT)<sup>64</sup>. The foreground list comprised ATAC-seq peaks at IMD loci showing colocalization only with caQTLs and not with eQTLs. The background list is all ATAC-seq peaks identified in LCLs that we tested for caQTL association.

## Relationship between peak-to-TSS distance and *cis*-heritability of caQTL and eQTL

The distance between the colocalized caQTL peak and the eGene's TSS (*i.e.*, peak-to-TSS distance) was determined to be the shortest distance from one end of the caQTL peak to the TSS. The pairs of caQTLs and eQTLs were split into quintiles based on their peak-to-TSS distance from closest to farthest.

MESC analysis uses REML implemented in GCTA<sup>102</sup> to estimate QTL *cis*-heritability. We referred to its output and compared the *cis*-heritability of caQTLs and eQTLs based on peak-to-TSS distance. To visualize the distribution of *cis*-heritability estimates, we grouped pairs of colocalized caQTLs and eQTLs based on peak-to-TSS distance quintiles.

## LCL eQTL meta-analysis

We downloaded LCL eQTL data from 3 studies through eQTL Catalogue release 6<sup>103</sup>. The sample sizes were 190, 147, and 418 for GENCORD<sup>69</sup>, GTEx<sup>55</sup>, and TwinsUK<sup>70</sup>, respectively. We meta-analyzed the summary statistics using the inverse variance weighted fixed effects model. If a

variant was missing in a subset of the studies, then only the available statistics were meta-analyzed. We used these meta-analyzed statistics to perform colocalization the same way as earlier colocalization analyses.

## Colocalization analysis with immune cell eQTL data

We downloaded eQTL data for various immune cell types from the eQTL Catalogue release 6<sup>103</sup>. The source studies from the eQTL Catalogue include BLUEPRINT<sup>73</sup>, DICE<sup>22</sup>, Alasoo *et al.*<sup>23</sup> and Bossini-Castillo *et al.*<sup>75</sup>. The represented immune cell types include T cell subtypes, B cells, neutrophils, and monocytes. We also separately downloaded data for CD4<sup>+</sup> T cells with and without stimulation<sup>74</sup>. The selection of candidate loci and colocalization analysis on them followed the same procedure as that for other QTLs.

We also evaluated whether meta-analysis of eQTL data can increase the number of loci with eQTL colocalization. We meta-analyzed eQTL data for 3 immune cell types with multiple sources – naïve CD4<sup>+</sup> T cell, monocyte, and memory regulatory T cell (Treg) – using the inverse variance weighted fixed effects model. Specifically, we meta-analyzed naïve CD4<sup>+</sup> T cell eQTL summary statistics from Soskic *et al.*<sup>74</sup>, BLUEPRINT<sup>73</sup>, and DICE<sup>22</sup>. We meta-analyzed monocyte eQTL summary statistics from BLUEPRINT<sup>73</sup> and DICE<sup>22</sup>. We meta-analyzed memory Treg eQTL summary statistics from Bossini-Castillo *et al.*<sup>75</sup> and DICE<sup>22</sup>. We used these meta-analyzed statistics to perform colocalization the same way as earlier colocalization analyses.

## Data availability

Processed data and code for generating the figures presented in the manuscript are available at <https://github.com/BulykLab/IMD-colocalization-manuscript-figures>. Genotype data are downloaded from the 1000 Genomes Project (<https://www.internationalgenome.org/data-portal/data-collection/30x-grch38>). LCL ATAC-seq data are downloaded from ENA (<https://www.ebi.ac.uk/ena/browser/view/PRJEB28318>). LCL RNA-seq count matrix is downloaded from Expression Atlas<sup>87</sup> (<https://www.ebi.ac.uk/gxa/experiments/E-GEUV-1/Results>). GWAS data for the studies listed in the Methods section are downloaded from the GWAS catalog<sup>104</sup> (<https://www.ebi.ac.uk/gwas/>). eQTL summary statistics are downloaded either from the eQTL Catalogue<sup>103</sup> (<https://www.ebi.ac.uk/eql/>) or the source publication.

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## Additional information

### Author contributions

R.J. and M.L.B. conceived and designed the research project. R.J. performed all analyses and prepared the figures. M.L.B. supervised the research. R.J. and M.L.B. wrote the manuscript.

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## Additional files

***Suppl Figs + Suppl Tables 5+7+10*** [🔗](#)

***Suppl Tables 1+2+3+4+6+8+9*** [🔗](#)

## References

1. Visscher P. M., et al. (2017) **10 Years of GWAS Discovery: Biology, Function, and Translation** *Am. J. Hum. Genet* **101**:5–22 [Google Scholar](#)
2. Claussnitzer M., et al. (2020) **A brief history of human disease genetics** *Nature* **577**:179–189 [Google Scholar](#)
3. Maurano M. T., et al. (2012) **Systematic localization of common disease-associated variation in regulatory DNA** *Science* **337**:1190–1195 [Google Scholar](#)
4. Gusev A., et al. (2014) **Partitioning heritability of regulatory and cell-type-specific variants across 11 common diseases** *Am. J. Hum. Genet* **95**:535–552 [Google Scholar](#)
5. Nicolae D. L., et al. (2010) **Trait-associated SNPs are more likely to be eQTLs: Annotation to enhance discovery from GWAS** *PLoS Genet* **6** [Google Scholar](#)
6. Hormozdiari F., et al. (2018) **Leveraging molecular quantitative trait loci to understand the genetic architecture of diseases and complex traits** *Nat. Genet* **50**:1041–1047 [Google Scholar](#)
7. O'Connor L. J., et al. (2019) **Extreme Polygenicity of Complex Traits Is Explained by Negative Selection** *Am. J. Hum. Genet* **105**:456–476 [Google Scholar](#)
8. Boyle E. A., Li Y. I., Pritchard J. K (2017) **An Expanded View of Complex Traits: From Polygenic to Omnigenic** *Cell* **169**:1177–1186 [Google Scholar](#)
9. Nasser J., et al. (2021) **Genome-wide enhancer maps link risk variants to disease genes** *Nature* **593**:238–243 [Google Scholar](#)
10. Weeks E. M., et al. (2023) **Leveraging polygenic enrichments of gene features to predict genes underlying complex traits and diseases** *Nat. Genet* **55**:1267–1276 [Google Scholar](#)
11. Morris J. A., et al. (2023) **Discovery of target genes and pathways at GWAS loci by pooled single-cell CRISPR screens** *Science* **380**:eadh7699 [Google Scholar](#)
12. Plenge R. M., Scolnick E. M., Altshuler D (2013) **Validating therapeutic targets through human genetics** *Nat. Rev. Drug Discov* **12**:581–594 [Google Scholar](#)
13. Chun S., et al. (2017) **Limited statistical evidence for shared genetic effects of eQTLs and autoimmune-disease-associated loci in three major immune-cell types** *Nat. Genet* **49**:600–605 [Google Scholar](#)
14. Barbeira A. N., et al. (2021) **Exploiting the GTEx resources to decipher the mechanisms at GWAS loci** *Genome Biol* **22** [Google Scholar](#)
15. Connally N. J., et al. (2022) **The missing link between genetic association and regulatory function** *eLife* **11** [Google Scholar](#)

16. Yao D. W., O'Connor L. J., Price A. L., Gusev A (2020) **Quantifying genetic effects on disease mediated by assayed gene expression levels** *Nat. Genet* :1–8 <https://doi.org/10.1038/s41588-020-0625-2> | [Google Scholar](#)
17. Umans B. D., Battle A., Gilad Y (2021) **Where Are the Disease-Associated eQTLs?** *Trends Genet* **37**:109–124 [Google Scholar](#)
18. Mostafavi H., Spence J. P., Naqvi S., Pritchard J. K (2023) **Systematic differences in discovery of genetic effects on gene expression and complex traits** *Nat. Genet* **55**:1866–1875 [Google Scholar](#)
19. Hukku A., et al. (2021) **Probabilistic Colocalization of Genetic Variants from Complex and Molecular Traits: Promise and Limitations** *Am. J. Hum. Genet* **108**:25–35 [Google Scholar](#)
20. Glassberg E. C., Gao Z., Harpak A., Lan X., Pritchard J. K (2019) **Evidence for weak selective constraint on human gene expression** *Genetics* **211**:757–772 [Google Scholar](#)
21. Dimas A. S., et al. (2009) **Common regulatory variation impacts gene expression in a cell type-dependent manner** *Science* **325**:1246–1250 [Google Scholar](#)
22. Schmiedel B. J., et al. (2018) **Impact of Genetic Polymorphisms on Human Immune Cell Gene Expression** *Cell* **175**:1701–1715 [Google Scholar](#)
23. Alasoo K., et al. (2018) **Shared genetic effects on chromatin and gene expression indicate a role for enhancer priming in immune response** *Nat. Genet* **50**:424–431 [Google Scholar](#)
24. Nathan A., et al. (2022) **Single-cell eQTL models reveal dynamic T cell state dependence of disease loci** *Nature* **606**:120–128 [Google Scholar](#)
25. Field A., Adelman K (2020) **Evaluating Enhancer Function and Transcription** *Annu. Rev. Biochem* **89**:213–234 [Google Scholar](#)
26. Kasowski M., et al. (2010) **Variation in transcription factor binding among humans** *Science* **328**:232–235 [Google Scholar](#)
27. Kilpinen H., et al. (2013) **Coordinated effects of sequence variation on DNA binding, chromatin structure, and transcription** *Science* **342**:744–747 [Google Scholar](#)
28. Waszak S. M., et al. (2015) **Population Variation and Genetic Control of Modular Chromatin Architecture in Humans** *Cell* **162**:1039–1050 [Google Scholar](#)
29. Degner J. F., et al. (2012) **DNase-I sensitivity QTLs are a major determinant of human expression variation** *Nature* **482**:390–394 [Google Scholar](#)
30. Kumasaka N., Knights A. J., Gaffney D. J (2019) **High-resolution genetic mapping of putative causal interactions between regions of open chromatin** *Nat. Genet* **51**:128–137 [Google Scholar](#)
31. Albert F. W., Kruglyak L (2015) **The role of regulatory variation in complex traits and disease** *Nat. Rev. Genet* **16**:197–212 [Google Scholar](#)
32. Wu Y., et al. (2023) **Joint analysis of GWAS and multi-omics QTL summary statistics reveals a large fraction of GWAS signals shared with molecular phenotypes** *Cell Genomics* **100344** <https://doi.org/10.1016/j.xgen.2023.100344> | [Google Scholar](#)

33. Jeong R., Bulyk M (2023) **L. Blood cell traits' GWAS loci colocalization with variation in PU.1 genomic occupancy prioritizes causal noncoding regulatory variants** *Cell genomics* **3**:100327 [Google Scholar](#)
34. Roadmap Epigenomics Consortium, et al. (2015) **Integrative analysis of 111 reference human epigenomes** *Nature* **518**:317–329 [Google Scholar](#)
35. Farh K. K. H., et al. (2015) **Genetic and epigenetic fine mapping of causal autoimmune disease variants** *Nature* **518**:337–343 [Google Scholar](#)
36. Giambartolomei C., et al. (2014) **Bayesian test for colocalisation between pairs of genetic association studies using summary statistics** *PLoS Genet* **10**:e1004383 [Google Scholar](#)
37. Pickrell J. K., et al. (2016) **Detection and interpretation of shared genetic influences on 42 human traits** *Nat. Genet* **48**:709–717 [Google Scholar](#)
38. Finucane H. K., et al. (2015) **Partitioning heritability by functional annotation using genome-wide association summary statistics** *Nat. Genet* **47**:1228–1235 [Google Scholar](#)
39. Cordell H. J., et al. (2021) **An international genome-wide meta-analysis of primary biliary cholangitis: Novel risk loci and candidate drugs** *J. Hepatol* **75**:572–581 [Google Scholar](#)
40. Dubois P. C. A., et al. (2010) **Multiple common variants for celiac disease influencing immune gene expression** *Nat. Genet* **42**:295–302 [Google Scholar](#)
41. de Lange K. M., et al. (2017) **Genome-wide association study implicates immune activation of multiple integrin genes in inflammatory bowel disease** *Nat. Genet* **49**:256–261 [Google Scholar](#)
42. López-Isac E., et al. (2021) **Combined genetic analysis of juvenile idiopathic arthritis clinical subtypes identifies novel risk loci, target genes and key regulatory mechanisms** *Ann. Rheum. Dis* **80**:321–328 [Google Scholar](#)
43. Patsopoulos N. A., et al. (2019) **Multiple sclerosis genomic map implicates peripheral immune cells and microglia in susceptibility** *Science* **365** [Google Scholar](#)
44. Ishigaki K., et al. (2022) **Multi-ancestry genome-wide association analyses identify novel genetic mechanisms in rheumatoid arthritis** *Nat. Genet* **54**:1640–1651 [Google Scholar](#)
45. Bentham J., et al. (2015) **Genetic association analyses implicate aberrant regulation of innate and adaptive immunity genes in the pathogenesis of systemic lupus erythematosus** *Nat. Genet* **47**:1457–1464 [Google Scholar](#)
46. Jin Y., et al. (2016) **Genome-wide association studies of autoimmune vitiligo identify 23 new risk loci and highlight key pathways and regulatory variants** *Nat. Genet* **48**:1418–1424 [Google Scholar](#)
47. Loh P. R., Kichaev G., Gazal S., Schoech A. P., Price A. L (2018) **Mixed-model association for biobank-scale datasets** *Nat. Genet* **50**:906–908 [Google Scholar](#)
48. Morris A. P., et al. (2012) **Large-scale association analysis provides insights into the genetic architecture and pathophysiology of type 2 diabetes** *Nat. Genet* **44**:981–990 [Google Scholar](#)

49. Schunkert H., et al. (2011) **Large-scale association analysis identifies 13 new susceptibility loci for coronary artery disease** *Nat. Genet* **43**:333–8 [Google Scholar](#)
50. Schizophrenia Working Group of the Psychiatric Genomics Consortium (2014) **Biological insights from 108 schizophrenia-associated genetic loci** *Nature* **511**:421–7 [Google Scholar](#)
51. van Mierlo H. C., Broen J. C. A., Kahn R. S., de Witte L. D (2019) **B-cells and schizophrenia: A promising link or a finding lost in translation?** *Brain. Behav. Immun* **81**:52–62 [Google Scholar](#)
52. Delaneau O., et al. (2019) **Chromatin three-dimensional interactions mediate genetic effects on gene expression** *Science* **364** [Google Scholar](#)
53. Heintzman N. D., et al. (2007) **Distinct and predictive chromatin signatures of transcriptional promoters and enhancers in the human genome** *Nat. Genet* **39**:311–318 [Google Scholar](#)
54. Project Consortium ENCODE, et al. (2020) **Expanded encyclopaedias of DNA elements in the human and mouse genomes** *Nature* **583**:699–710 [Google Scholar](#)
55. GTEx Consortium (2020) **The GTEx Consortium atlas of genetic regulatory effects across human tissues** *Science* **369**:1318–1330 [Google Scholar](#)
56. Lappalainen T., et al. (2013) **Transcriptome and genome sequencing uncovers functional variation in humans** *Nature* **501**:506–511 [Google Scholar](#)
57. Kundu K., et al. (2022) **Genetic associations at regulatory phenotypes improve fine-mapping of causal variants for twelve immune-mediated diseases** *Nat. Genet* **440** [Google Scholar](#)
58. Liu T., et al. (2011) **Cistrome: an integrative platform for transcriptional regulation studies** *Genome Biol* **12**:R83 [Google Scholar](#)
59. Thomas P. D., et al. (2022) **PANTHER: Making genome-scale phylogenetics accessible to all** *Protein Sci* **31**:8–22 [Google Scholar](#)
60. Sanmartí R., Ruiz-Esquide V., Bastida C., Soy D (2018) **Tocilizumab in the treatment of adult rheumatoid arthritis** *Immunotherapy* **10**:447–464 [Google Scholar](#)
61. Khanna R., Feagan B. G (2013) **Ustekinumab for the treatment of Crohn’s disease** *Immunotherapy* **5**:803–15 [Google Scholar](#)
62. Monemi S., et al. (2016) **Incidence of Gastrointestinal Perforations in Patients with Rheumatoid Arthritis Treated with Tocilizumab from Clinical Trial, Postmarketing, and Real-World Data Sources** *Rheumatol. Ther* **3**:337–352 [Google Scholar](#)
63. Stevenson C., et al. (2014) **Essential role of Elmo1 in Dock2-dependent lymphocyte migration** *J. Immunol* **192**:6062–70 [Google Scholar](#)
64. McLean C. Y., et al. (2010) **GREAT improves functional interpretation of cis-regulatory regions** *Nat. Biotechnol* **28**:495–501 [Google Scholar](#)

65. Tanigawa Y., Dyer E. S., Bejerano G (2022) **WhichTF is functionally important in your open chromatin data?** *PLoS Comput. Biol* **18**:e1010378 [Google Scholar](#)
66. Fulco C. P., et al. (2019) **Activity-by-contact model of enhancer-promoter regulation from thousands of CRISPR perturbations** *Nat. Genet* **51** [Google Scholar](#)
67. Zuin J., et al. (2022) **Nonlinear control of transcription through enhancer-promoter interactions** *Nature* **604**:571–577 [Google Scholar](#)
68. Vösa U., et al. (2021) **Large-scale cis- and trans-eQTL analyses identify thousands of genetic loci and polygenic scores that regulate blood gene expression** *Nat. Genet* **53** [Google Scholar](#)
69. Gutierrez-Arcelus M., et al. (2013) **Passive and active DNA methylation and the interplay with genetic variation in gene regulation** *eLife* **2013**:1–18 [Google Scholar](#)
70. Buil A., et al. (2015) **Gene-gene and gene-environment interactions detected by transcriptome sequence analysis in twins** *Nat. Genet* **47**:88–91 [Google Scholar](#)
71. Dziembowska M., et al. (2002) **Three novel mutations of the CIITA gene in MHC class II-deficient patients with a severe immunodeficiency** *Immunogenetics* **53**:821–9 [Google Scholar](#)
72. Hofmann E., et al. (2010) **Octamer-binding factor 6 (Oct-6/Pou3f1) is induced by interferon and contributes to dsRNA-mediated transcriptional responses** *BMC Cell Biol* **11** [Google Scholar](#)
73. Chen L., et al. (2016) **Genetic Drivers of Epigenetic and Transcriptional Variation in Human Immune Cells** *Cell* **167**:1398–1414 [Google Scholar](#)
74. Soskic B., et al. (2022) **Immune disease risk variants regulate gene expression dynamics during CD4+ T cell activation** *Nat. Genet* **54**:817–826 [Google Scholar](#)
75. Bossini-Castillo L., et al. (2022) **Immune disease variants modulate gene expression in regulatory CD4+ T cells** *Cell Genomics* **2** [Google Scholar](#)
76. Xu W.-D., Li R., Huang A.-F (2022) **Role of TL1A in Inflammatory Autoimmune Diseases: A Comprehensive Review** *Front. Immunol* **13** [Google Scholar](#)
77. Wang X., Goldstein D. B (2020) **Enhancer Domains Predict Gene Pathogenicity and Inform Gene Discovery in Complex Disease** *Am. J. Hum. Genet* **106**:215–233 [Google Scholar](#)
78. Kim S., Wysocka J (2023) **Deciphering the multi-scale, quantitative cis-regulatory code** *Mol. Cell* **83**:373–392 [Google Scholar](#)
79. Turley P., et al. (2018) **Multi-trait analysis of genome-wide association summary statistics using MTAG** *Nat. Genet* **50**:229–237 [Google Scholar](#)
80. Danecek P., et al. (2021) **Twelve years of SAMtools and BCFtools** *Gigascience* **10** [Google Scholar](#)



81. Quinlan A. R., Hall I. M (2010) **BEDTools: a flexible suite of utilities for comparing genomic features** *Bioinformatics* **26**:841–2 [Google Scholar](#)
82. Langmead B., Salzberg S. L (2012) **Fast gapped-read alignment with Bowtie 2** *Nat. Methods* **9**:357–359 [Google Scholar](#)
83. Dunham I., et al. (2012) **An integrated encyclopedia of DNA elements in the human genome** *Nature* **489**:57–74 [Google Scholar](#)
84. Van De Geijn B., Mcvicker G., Gilad Y., Pritchard J. K (2015) **WASP: Allele-specific software for robust molecular quantitative trait locus discovery** *Nat. Methods* **12**:1061–1063 [Google Scholar](#)
85. Zhang Y., et al. (2008) **Model-based analysis of ChIP-Seq (MACS)** *Genome Biol* **9** [Google Scholar](#)
86. Banovich N. E., et al. (2018) **Impact of regulatory variation across human iPSCs and differentiated cells** *Genome Res* **28**:122–131 [Google Scholar](#)
87. Papatheodorou I., et al. (2020) **Expression Atlas update: from tissues to single cells** *Nucleic Acids Res* **48**:D77–D83 [Google Scholar](#)
88. Byrska-Bishop M., et al. (2022) **High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios** *Cell* **185**:3426–3440 [Google Scholar](#)
89. Auton A., et al. (2015) **A global reference for human genetic variation** *Nature* **526**:68–74 [Google Scholar](#)
90. Das S., et al. (2016) **Next-generation genotype imputation service and methods** *Nat. Genet* **48**:1284–1287 [Google Scholar](#)
91. Browning B. L., Zhou Y., Browning S. R (2018) **A One-Penny Imputed Genome from Next-Generation Reference Panels** *Am. J. Hum. Genet* **103**:338–348 [Google Scholar](#)
92. Browning B. L., Tian X., Zhou Y., Browning S. R (2021) **Fast two-stage phasing of large-scale sequence data** *Am. J. Hum. Genet* **108**:1880–1890 [Google Scholar](#)
93. Cortes A., Brown M. A (2011) **Promise and pitfalls of the Immunochip** *Arthritis Res. Ther* **13** [Google Scholar](#)
94. Kent W. J., et al. (2002) **The human genome browser at UCSC** *Genome Res* **12**:996–1006 [Google Scholar](#)
95. Lyon M. S., et al. (2021) **The variant call format provides efficient and robust storage of GWAS summary statistics** *Genome Biol* **22** [Google Scholar](#)
96. Liao Y., Smyth G. K., Shi W (2014) **featureCounts: an efficient general purpose program for assigning sequence reads to genomic features** *Bioinformatics* **30**:923–30 [Google Scholar](#)
97. Robinson M. D., Oshlack A (2010) **A scaling normalization method for differential expression analysis of RNA-seq data** *Genome Biol* **11** [Google Scholar](#)
98. Buja A., Eyuboglu N (1992) **Remarks on Parallel Analysis** *Multivariate Behav. Res* **27**:509–40 [Google Scholar](#)

99. Zhou H. J., Li L., Li Y., Li W., Li J. J. (2022) **PCA outperforms popular hidden variable inference methods for molecular QTL mapping** *Genome Biol* **23** [Google Scholar](#)
100. Taylor-Weiner A., et al. (2019) **Scaling computational genomics to millions of individuals with GPUs** *Genome Biol* **20** [Google Scholar](#)
101. Mi H., et al. (2019) **Protocol Update for large-scale genome and gene function analysis with the PANTHER classification system (v.14.0)** *Nat. Protoc* **14**:703–721 [Google Scholar](#)
102. Yang J., Lee S. H., Goddard M. E., Visscher P. M. (2011) **GCTA: a tool for genome-wide complex trait analysis** *Am. J. Hum. Genet* **88**:76–82 [Google Scholar](#)
103. Kerimov N., et al. (2021) **A compendium of uniformly processed human gene expression and splicing quantitative trait loci** *Nat. Genet* **53**:1290–1299 [Google Scholar](#)
104. Sollis E., et al. (2023) **The NHGRI-EBI GWAS Catalog: knowledgebase and deposition resource** *Nucleic Acids Res* **51**:D977–D985 [Google Scholar](#)

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## Editors

Reviewing Editor

### Murim Choi

Seoul National University, Seoul, Republic of Korea

Senior Editor

### Murim Choi

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### Reviewer #1 (Public review):

Most human traits and common diseases are polygenic, influenced by numerous genetic variants across the genome. These variants are typically non-coding and likely function through gene regulatory mechanisms. To identify their target genes, one strategy is to examine if these variants are also found among genetic variants with detectable effects on

gene expression levels, known as eQTLs. Surprisingly, this strategy has had limited success, and most disease variants are not identified as eQTLs, a puzzling observation recently referred to as "missing regulation".

In this work, Jeong and Bulyk aimed to better understand the reasons behind the gap between disease-associated variants and eQTLs. They focused on immune-related diseases and used lymphoblastoid cell lines (LCLs) as a surrogate for the cell types mediating the genetic effects. Their main hypothesis is that some variants without eQTL evidence might be identifiable by studying other molecular intermediates along the path from genotype to phenotype. They specifically focused on variants that affect chromatin accessibility, known as caQTLs, as a potential marker of regulatory activity.

The authors present data analyses supporting this hypothesis: several disease-associated variants are explained by caQTLs but not eQTLs. They further show that although caQTLs and eQTLs likely have largely overlapping underlying genetic variants, some variants are discovered only through one of these mapping strategies. Notably, they demonstrate that eQTL mapping is underpowered for gene-distal variants with small effects on gene expression, whereas caQTL mapping is not dependent on the distance to genes. Additionally, for some disease variants with caQTLs but no corresponding eQTLs in LCLs, they identify eQTLs in other cell types.

Altogether, Jeong and Bulyk convincingly demonstrate that for immune-related diseases, discovering the missing disease-eQTLs requires both larger eQTL studies and a broader range of cell types in expression assays. It remains to be seen what fractions of the missing disease-eQTLs will be discovered with either strategy and whether these results can be extended to other diseases or traits.

It should be noted that the problem of "missing regulation" has been investigated and discussed in several recent papers, notably Umans et al., *Trends in Genetics* 2021; Connally et al., *eLife* 2022; Mostafavi et al., *Nat. Genet.* 2023. The results reported by Jeong and Bulyk are not unexpected in light of this previous work (all of which they cite), but they add valuable empirical evidence that mostly aligns with the model and discussions presented in Mostafavi et al.

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#### **Reviewer #2 (Public review):**

eQTLs have emerged as a method for interpreting GWAS signals. However, some GWAS signals are difficult to explain with eQTLs. In this paper, the authors demonstrated that caQTLs can explain these signals. This suggests that for GWAS signals to actually lead to disease phenotypes, they must be accessible in the chromatin. This implies that for GWAS signals to translate into disease phenotypes, they need to be accessible within the chromatin.

However, fundamentally, caQTLs, like GWAS, have the limitation of not being able to determine which genes mediate the influence on disease phenotypes. This limitation is consistent with the constraints observed in this study.

(1) Reproducibility / Methods. The concrete numbers provided in the authors' response (e.g., 20 YRI LCL ATAC-seq samples used only for peak discovery; caQTL mapping restricted to 100 GBR LCLs; 99,320 ATAC peaks tested vs 14,872 genes for eQTL; 373 European RNA-seq samples, with clarification of overlap) do not appear to be reflected in the Methods. These specifics should be incorporated directly into the Methods sections.

(2) Experimental evidence demonstrating transcription factor binding at representative caQTL peaks would strengthen causal interpretation of these loci.

(3) Tissue/cell-type specificity of caQTLs: Prior work supports that chromatin-level effects are broadly shared across cellular states, whereas expression effects are more context-specific; thus, caQTLs are generally less "state-specific" than eQTLs. However, this does not imply equivalence across distinct cell types: caQTLs derived from different cell types may yield different results, particularly where accessibility is cell-type restricted.

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### Author response:

The following is the authors' response to the original reviews.

#### **Reviewer #1 (Public Review):**

*Most human traits and common diseases are polygenic, influenced by numerous genetic variants across the genome. These variants are typically non-coding and likely function through gene regulatory mechanisms. To identify their target genes, one strategy is to examine if these variants are also found among genetic variants with detectable effects on gene expression levels, known as eQTLs. Surprisingly, this strategy has had limited success, and most disease variants are not identified as eQTLs, a puzzling observation recently referred to as "missing regulation".*

*In this work, Jeong and Bulyk aimed to better understand the reasons behind the gap between disease-associated variants and eQTLs. They focused on immune-related diseases and used lymphoblastoid cell lines (LCLs) as a surrogate for the cell types mediating the genetic effects. Their main hypothesis is that some variants without eQTL evidence might be identifiable by studying other molecular intermediates along the path from genotype to phenotype. They specifically focused on variants that affect chromatin accessibility, known as caQTLs, as a potential marker of regulatory activity.*

*The authors present data analyses supporting this hypothesis: several disease-associated variants are explained by caQTLs but not eQTLs. They further show that although caQTLs and eQTLs likely have largely overlapping underlying genetic variants, some variants are discovered only through one of these mapping strategies. Notably, they demonstrate that eQTL mapping is underpowered for gene-distal variants with small effects on gene expression, whereas caQTL mapping is not dependent on the distance to genes. Additionally, for some disease variants with caQTLs but no corresponding eQTLs in LCLs, they identify eQTLs in other cell types.*

*Altogether, Jeong and Bulyk convincingly demonstrate that for immune-related diseases, discovering the missing disease-eQTLs requires both larger eQTL studies and a broader range of cell types in expression assays. It remains to be seen what fractions of the missing disease-eQTLs will be discovered with either strategy and whether these results can be extended to other diseases or traits.*

We thank the reviewer for their accurate summary of our study and positive review of our findings for immune-related diseases.

*It should be noted that the problem of "missing regulation" has been investigated and discussed in several recent papers, notably Umans et al., Trends in Genetics 2021; Connally et al., eLife 2022; Mostafavi et al., Nat. Genet. 2023. The results reported by Jeong and Bulyk are not unexpected in light of this previous work (all of which they cite), but they add valuable empirical evidence that mostly aligns with the model and discussions presented in Mostafavi et al.*

We thank the reviewer for their positive review of our results and manuscript. As Reviewer #1 noted, whether our and others' observation extends to other diseases or traits is an open question. For instance, Figure 2b in Mostafavi et al., Nat. Genet. (2023) demonstrated that there was a spectrum of depletion of eQTLs and enrichment of GWAS signals in constrained genes across various tissues and traits, respectively. Therefore, gene expression constraint may play a larger or smaller role in different diseases or traits. That immune cell types and cell states are extremely diverse (Schmiedel et al., Cell (2018) and Calderon et al., Nat. Genet. (2019), just to name a few) likely adds to the complexity of gene regulation that contributes to immune-mediated disease.

**Reviewer #2 (Public Review):**

*Summary:*

*eQTLs have emerged as a method for interpreting GWAS signals. However, some GWAS signals are difficult to explain with eQTLs. In this paper, the authors demonstrated that caQTLs can explain these signals. This suggests that for GWAS signals to actually lead to disease phenotypes, they must be accessible in the chromatin. This implies that for GWAS signals to translate into disease phenotypes, they need to be accessible within the chromatin.*

*However, fundamentally, caQTLs, like GWAS, have the limitation of not being able to determine which genes mediate the influence on disease phenotypes. This limitation is consistent with the constraints observed in this study.*

We thank the reviewer for their accurate summary of our results.

*(1) For reproducibility, details are necessary in the method section.*

*Details about adding YRI samples in ATAC-seq: For example, how many samples are there, and what is used among public data? There is LCL-derived iPSC and differentiated iPSC (cardiomyocytes) data, not LCL itself. How does this differ from LCL, and what is the rationale for including this data despite the differences?*

Banovich et al., Genome Research (2018) (PMID: 29208628), who generated data using LCL-derived iPSCs and differentiated iPSCs (cardiomyocytes), also generated ATAC-seq data from 20 YRI LCL samples. We analyzed those data to identify open chromatin regions (i.e., ATACseq peaks) in LCLs and merged the regions with open chromatin regions identified with 100 GBR LCL samples from two studies by Kumasaka et al. (Nature Genetics (2016)

PMID: 26656845 and Nature Genetics (2019) PMID: 30478436). However, we restricted the caQTL analysis to only the 100 GBR samples because of possible ancestry effects and batch effects. We attempted caQTL analysis with the 20 YRI samples as well, but the result was noisy, likely due to smaller sample size and lower read depth of the ATAC-seq data.

*caQTL is described as having better power than eQTL despite having fewer samples. How does the number of ATAC peaks used in caQTL compare to the number of gene expressions used in eQTL?*

The number of ATAC peaks used in caQTL (99,320) is ~6.7 times greater than the number of genes (14,872) used in the eQTL analysis. Therefore, there is a higher chance of detecting a significant caQTL signal and a significant colocalization signal than there is for eQTLs. However, we reasoned that since distal eQTLs are more easily detected as caQTLs and since increasing the sample size of eQTLs through meta-analysis uncovered additional eQTL

colocalization at loci with caQTL colocalization only, colocalized caQTLs are likely capturing disease-relevant regulatory effects.

*Details about RNA expression data: In the method section, it states that raw data (ERP001942) was accessed, and in data availability, processed data (E-GEUV-1) was used. These need to be consistent.*

Thank you for pointing this out. We used the processed data from Expression Atlas (<https://www.ebi.ac.uk/gxa/experiments/E-GEUV-1/Results>), and that's what we meant by "We downloaded RNA expression level data of the LCL samples from the Expression Atlas." We have revised the "RNA expression data preparation" section in our manuscript to make the text clearer.

*How many samples were used (the text states 373, but how was it reduced from the original 465, and the total genotype is said to be 493 samples while ATAC has n=100; what are the 20 others?), and it mentions European samples, but does this exclude YRI?*

We thank the reviewer for pointing out these points of confusion. Our reported count of 493 samples included YRI samples with RNA-seq data or ATAC-seq data that we ultimately did not use for QTL analyses. There were 373 European samples with RNA-seq data that we used for eQTL analysis, and 100 GBR samples (including some that overlap with the 373 European samples) that we used for caQTL analysis. We have revised the text to clarify these points.

*(2) Experimental results determining which TFs might bind to the representative signals of caQTL are required.*

We agree that caQTL colocalization is just the start of elucidating the regulatory mechanism of a GWAS locus. Determining which TFs are bound and which TFs' binding is altered would be necessary to describe the causal regulatory mechanism. For this, we utilized the Cistrome database to search for TFs whose binding overlaps the colocalized caQTL peaks. We present the results of this analysis in Supplementary Table 3 and Supplementary Figure 4, both of which we have added in our revised manuscript. Overall, protein factors associated with active transcription, such as POL2RA, and several immune cell TFs, including RUNX3, SPI1, and RELA, were frequently detected in those peaks. Detecting these factors in most peaks supports the likelihood that the colocalized caQTL peaks are active cis-regulatory elements. These results are consistent with our observation of enriched caQTL-mediated heritability in regions with active histone marks (Figure 1).

*(3) It is stated that caQTL is less tissue-specific compared to eQTL; would caQTL performed with ATAC-seq results from different cell types, yield similar results?*

We thank the reviewer for the question. Calderon et al. (PMID: 31570894) observed that "most effects on allelic imbalance (of ATAC-seq) were shared regardless of lineage or condition". Yet, there were regions where a different cell type or state would show inaccessibility (Figure 4d in Calderon et al.). Thus, we expect that ATAC-seq results from different cell types (e.g., T cells, B cells, monocytes, etc.) would lead to additional caQTLs showing colocalization at cell-typespecific open chromatin. However, if a region is accessible in both cell types, caQTL may be detected in both. Moreover, Alasoo et al., Nature Genetics (2018) (PMID: 29379200) observed that "many disease-risk variants affect chromatin structure in a broad range of cellular states, but their effects on expression are highly context specific." In both studies, the authors investigated immune cell types, and there could be different observations in non-immune cell types and other diseases and traits.



### Reviewer #1 (Recommendations For The Authors):

*I think it would strengthen the paper to explore gene-level differences in the discovery of caQTLs and eQTLs. For example, complex disease-relevant genes, on average, have more/longer regulatory domains (as shown by Wang and Goldstein, AJHG 2020; Mostafavi et al., Nat. Genet. 2023). Therefore, it is plausible that for such genes, caQTLs are much more easily discoverable than eQTLs due to (i) a larger mutational target size for caQTLs, and (ii) dispersion of expression heritability across multiple domains, which hampers the discovery of eQTLs but not caQTLs, which are studied independently of other domains in the region. In other words, discovered caQTLs and eQTLs likely vary in terms of their distance to genes (as the authors report), as well as their target genes.*

We thank the reviewer for the suggestion to explore gene-level differences. We expect that the effects of complex disease-relevant genes having more / longer regulatory domains, on average, to explain our observations. We agree on both of your points that there are many more regulatory elements that are captured as accessible regions than expressed genes and that genes often have multiple independent eQTLs leading to dispersion of heritability. The genelevel trend that we described was the distance of the regulatory element from the genes. Additional analyses would be a relevant future direction.

*Also considering gene-level analysis, Mostafavi et al. show that the types of biases they report for eQTLs also apply to other molecular QTLs. It would be valuable to compare GWAS hits with versus without caQTL colocalization. Similarly, it would be insightful to compare GWAS hits with both colocalized caQTLs and eQTLs to GWAS hits with colocalized caQTLs but no eQTLs in any of the cell types.*

We thank the reviewer for the comment. Investigating for potential biases in the colocalized caQTL would be useful, but we considered it beyond the scope of this work. In terms of biological factors, we demonstrated through mediated heritability analyses that more accessible chromatin (based on ATAC-seq read coverage) and regions with active histone marks were enriched for autoimmune disease associations (Figure 1). Furthermore, as greater distance of the regulatory variant from the transcription start site significantly reduced the cis-heritability, we would expect that distance would play a major role, similar to Mostafavi et al.'s conclusions.

*I don't think the argument for the role of natural selection contributing to the "missing regulation" is presented accurately. Specifically, large eQTLs acting on top trait-relevant genes are under stronger selection and thus, on average, segregate at lower frequencies. This makes them difficult to discover in eQTL assays. However, if not lost, they contribute as much, if not more, to trait heritability than weaker eQTLs at the same gene because their larger effects compensate for their lower frequency. At the most extreme, selection should have a "flattening" effect (e.g., see Simons et al., PLOS Biol 2018; O'Connor et al., AJHG 2019): weak and strong eQTLs at the same gene are expected to contribute equally to heritability. Therefore, the statement "Consequently, only weak eQTL variants, often in regions distal to the gene's promoter, may remain and affect traits" is not correct. If this turns out to be empirically true, other models, such as pleiotropic selection, need to explain it.*

We thank the reviewer for the correction. We agree with the comment and have revised the sentences in the introduction accordingly.

*It is worth speculating why caQTLs may be more consistent across cell types than cis-eQTLs. Additionally, readers may infer from the paper that the focus should shift from eQTLs to caQTLs, which may not be the authors' intention. Perhaps these approaches are*

*complementary: caQTLs can help with TSS-distal disease variants, while finding the target gene and regulatory context is more straightforward with eQTL colocalization. Addressing these points in the discussion will be helpful.*

We appreciate the reviewer's suggestion to clarify the advantages of incorporating cis-eQTLs and caQTLs. Our argument is exactly as you put it, and we added a paragraph on this in the Discussion.

*I believe the authors could do more to contextualize their findings within the existing literature on the subject, particularly Umans et al., Trends in Genetics 2021; Connally et al., eLife 2022; and Mostafavi et al., Nat. Genet. 2023. For instance, Umans et al. suggest that "if most standard eQTLs are generally benign, increasing sample size and adding more tissue types in an effort to identify even more standard eQTLs may not help us to explain many more disease risk mutations". Conversely, Mostafavi et al. argue for a multipronged approach, which appears more aligned with the authors' conclusions.*

We followed the reviewer's suggestion to place our work in the context of existing literature on this topic. Moreover, we clarified what our recommendations for future data generation are.

*I thought Figures 1C-D were unclear.*

We added a sentence in the figure legend describing that stronger and more significant enrichment indicate that mediated heritability is concentrated in that subset.

**Reviewer #2 (Recommendations For The Authors):**

*Complete workflow figures for caQTL calling and eQTL calling are required.*

To improve clarity of the caQTL and eQTL calling workflow, we added Supplementary Figure 1.

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